

Scaling Laws, Class Invariance, and the Limits of Audibility

of an Explicit Prime–Zero Coupling Operator

Ulrich Tehrani

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Abstract

This paper determines what a prime-indexed coupling operator can and cannot hear. The operator $\tilde{T}(\sigma) = \Phi(\sigma)\Phi(\sigma)^*$ of the present series is built explicitly from prime phasors; no spectral property is postulated. Three questions are settled about it, and the order in which they are settled matters. A symmetry barrier comes first: every symmetric functional of the orbit of a zero under the functional equation is even in the signed distance from the critical line, so the class of observables studied here is blind to the left/right orientation of that displacement. Evenness needs no regularity; under differentiability the first-order response vanishes, and for the curvature sums of this series, whose dependence enters through even orbit averages of the entire weighted curvature test kernel, the first possible non-constant term is quadratic; it is the first surviving term unless the aggregate quadratic coefficient cancels. This is a theorem, not a limitation of effort. What remains is the audible part — audibility always meaning separation from the stated comparison worlds by the stated observable class — and it splits cleanly. On the prime side, what is audible is stated exactly as certified: the prime-side weighted point configuration is compared with three control objects kept separate throughout — a smooth measure whose own zeta function is zero-free, its rescaling to the exact mass of the prime counting measure, and a discrete equal-mass quadrature world. Certified separation margins are established against the smooth measure and the discrete world; only the rescaling matches the prime count exactly; and for the discrete pair no functional of the counting data with bounded-Lipschitz constant below an explicit threshold factors the separation. This identifies the point configuration relative to the stated controls, not multiplicativity as its cause: the controls vary point geometry and multiplicative structure together. On the prime input side, the scaling laws are invariant across the Beurling class: every admissible system carries the same limiting function, the same certified separation and the same leading profile with the same numerical band — a within-class non-distinction, proved, not a separation. On the ordinate side, a pre-registered unfolded discrimination test over five worlds and five observables finds no separation beyond the imposed unit-density skeleton, and the scope of that measurement is declared.

What is proved. The kernel identity representing the companion prime-space Gram operator $T = \Phi^*\Phi$ as the damped inverse spectrum evaluated at prime ratios and products (Theorem 2.2); canonicity of the phasor kernel within the unimodular trigonometric almost periodic class (Lemma 2.4, Proposition 2.5); non-convergence of the normalised leading spectral ratio at the reference data (100, 0.05), with a certified two-point separation (Theorem 3.1); divergence of the mean energy-asymmetry functional along a bounded leading

profile (Theorem 3.4); the ordinate tail bound, which shows that the ordinate count is a truncation and not a limit parameter (Theorem 3.5); transfer of both scaling laws to Beurling systems, carrying the same limiting objects and the same leading profile class-independently (Theorem 4.2) together with the sideband transfer (Theorem 4.4); the summation formula for the smooth measure (Theorem 5.2), the quadrature estimate for the discrete world (Theorem 5.3) and the factorisation barrier (Theorem 5.5); and the symmetry barrier (Theorem 6.1).

What is certified. The two-point separation of the limiting ratio; the discrimination margins in curvature against the smooth control world, uniformly on the reference window; and the bounded-Lipschitz distance between the two weighted counting measures. Each certified statement names its verification protocol where it is first stated.

What is numerical. The outcome of the pre-registered unfolded discrimination test (§ 6), reported as a measurement with its design, its criterion, its result and its scope; the completeness counts underlying it; the embedding of the sideband transfer in an explicit test world; and the robustness of the second moment under completion of the mode set to prime powers (Remark 2.2); and a function-side reconstruction experiment, separate from the battery, reported with its design, its metrics and its scope.

What remains open. Pointwise, as opposed to mean, finality of the divergence (§ 9); the minimal order break that destroys the class laws; the exceptional-set measure under intermediate conditions; extension of the factorisation barrier from one pair of worlds to families; the identification design that would attribute the certified prime-side separation to a single feature; pair and correlation statistics on the unfolded ordinate axis; and the combinatorial gap that separates the canonicity lemma from the full Bohr class.

Structure. The formal core is §§ 2–6, and it is non-circular: every statement there carries its status in its heading — proved, certified, or proved from certified inputs — and numerical observations are labelled as such where they occur. No use of the Riemann Hypothesis, the GUE conjecture, or the Hilbert–Pólya postulate is made anywhere in §§ 2–6.

Notation and Conventions

Critical line. Throughout, σ denotes $\Re(s)$ and the critical line means $\sigma = \frac{1}{2}$. The trace parameter of the operator family is likewise written σ and is evaluated at $\sigma_0 = \frac{1}{2}$ unless stated otherwise.

Global parameters. The reference values are $\kappa = 53$ (prime cutoff, giving 16 active primes $p \leq \kappa$), $\varepsilon = 0.05$ (Gaussian regularisation parameter), and $N = 100$ (number of Riemann ordinates γ_k employed). Certified statements are given on the window $\varepsilon \in [0.04, 0.07]$. The unfolded scale used in § 6 is $\tilde{\varepsilon} = 0.25$; it is derived, not chosen, from the effective bandwidth at the reference parameters.

Main objects. The coupling map $\Phi(\sigma) : H_{\text{str}} \rightarrow H_{\text{null}}$ between the finite-dimensional prime space $H_{\text{str}} = \ell^2(\{p \leq \kappa\})$ and the ordinate space $H_{\text{null}} = \ell^2(\{\gamma_1, \dots, \gamma_N\})$ is

$$\Phi(\sigma)_{k,p} = e^{-\varepsilon^2 \gamma_k^2 / 2} \sin(\sigma \gamma_k \log p),$$

and the coupling operator is $\tilde{T}(\sigma) = \Phi(\sigma) \circ \Phi(\sigma)^*$ on H_{null} , with loop operator $T(\sigma) = \Phi(\sigma)^* \circ$

$\Phi(\sigma)$ on H_{str} . The oscillatory trace sum and the curvature sum are

$$O(\sigma) = \frac{1}{2} \sum_{k,p} e^{-\varepsilon^2 \gamma_k^2} \cos(2\sigma \gamma_k \log p), \quad B = \sum_{k,p} e^{-\varepsilon^2 \gamma_k^2} (\gamma_k \log p)^2 \cos(\gamma_k \log p),$$

the second being the on-line curvature sum with $O''(\frac{1}{2}) = -2B$. The damped inverse spectrum entering § 2 is

$$G(u) = \sum_{k=1}^N e^{-\varepsilon^2 \gamma_k^2} \cos(\sigma \gamma_k u),$$

a single even function of one real variable. The Beurling density class of § 4 is

$$\mathcal{D}_\delta = \{\mathfrak{B} : \vartheta_{\mathfrak{B}}(x) = x + O(x/(\log x)^{2+\delta})\}, \quad \delta > 0.$$

Weights. Two prime weights occur in this series and are numerically distinct. With

$$V_p(\tfrac{1}{2}) = \frac{4(\log p)^2 p}{(p-1)^2}, \quad f_p = V_p(\tfrac{1}{2}) - \frac{4(\log p)^2}{p} = \frac{4(\log p)^2 (2p-1)}{p(p-1)^2},$$

the η -framework weight is $c_p^\eta := \sqrt{V_p(\frac{1}{2})}$ and the renormalised Weil weight is $c_p^{\text{ren}} := \sqrt{f_p}$. At the reference parameters the associated values of the energy-asymmetry functional are $\eta_{\text{orig}}(\frac{1}{2}) = 0.69078176$ and $\eta_{\text{ren}}(\frac{1}{2}) = 0.66926873$. In the convention of the earlier papers $\eta[c] = 1 - E_{\text{spec}}/E_{\text{str}}$, so that $E_{\text{spec}}/E_{\text{str}} = 1 - \eta$ is the Rayleigh quotient of that series; $\eta \rightarrow -\infty$ therefore means $E_{\text{spec}}/E_{\text{str}} \rightarrow +\infty$. Both are evaluated in the convention in which the phasor argument carries no trace parameter, that is $\sin(\gamma_k \log p)$; this is the convention of the earlier papers of the series in which these quantities were introduced, and it does not coincide with $\sin(\sigma \gamma_k \log p)$ at $\sigma = \frac{1}{2}$.

Analytical foundations. The proofs use the prime number theorem with classical error term together with Mertens' theorem, Weyl's eigenvalue perturbation inequality, Dirichlet's pigeon-hole principle, and, at exactly one place, the two-function Parseval identity for Besicovitch B^2 almost periodic functions. The numerical completeness checks of § 6 use the argument principle; it enters no proof.

Numerical computations. Working precisions vary by artefact and are declared here: the five-world battery runs at 40 digits with `float64` observable paths; the function-side reconstruction at 8 digits with NumPy/SciPy; the two margin and separation certificates in Arb ball arithmetic at 192 and 320 bits through `python-flint`, with an `mpmath` cross-layer at 32 digits. Riemann ordinates γ_k enter through the normative layer of § 8, regenerated via `mpmath.zetazero` or isolated in Arb, in either case under exact double-precision cross-check. Verification code is available at <https://github.com/utehrani/analysislab-nt>.

Not to be confused with. The following pairs are distinct objects and are never interchanged.

Symbol	Meaning	Distinct from
$G(u)$	damped inverse spectrum	g , the candidate kernel of § 2
$\mathcal{F}$	Bohr spectrum of g	$\Lambda(n)$, von Mangoldt
ω_0	unimodular constant phase	$\zeta(s)$
$\mathcal{O}$	observable (functional on worlds)	$O(\sigma)$, trace sum
$D_{\text{SEL}} \approx 10.9845$	trace-formula constant	$D \approx 9.470$
c_p^η	η -framework weight	c_p^{ren} , Weil weight
B	curvature sum, γ_k^2 weights	$B_{\text{int}}^+(\varepsilon, N)$, integrated bias
$r(x)$	normalised leading spectral ratio	bias ratios r_p of the earlier papers
$\mathcal{K}_{\kappa, \varepsilon}(z)$	aggregated curvature test kernel	$G(u)$ (ordinate sum); K_ε (§ 5)
φ, K_ε	test function, curvature kernel	g , the candidate kernel of § 2
$\mathfrak{B}$ (§ 4)	Beurling system index	$B, B_{\text{sm}}, B[A], B_{\text{int}}^+$; world A (§ 5)

The curvature sums attached to different point sets are distinguished throughout by their argument: $B[P]$ for the primes $p \leq \kappa$, $B[A]$ for the discrete quadrature world A , and B_{sm} for the smooth measure μ_{sm} of § 5. The three control objects of that section — μ_{sm} , its mass-matched rescaling μ_{norm} , and the discrete world A — are kept apart throughout and are never referred to collectively as “the control world”.

1 Introduction

1.1 Background and Motivation

This paper is the ninth in a series developing an operator-theoretic framework for studying the distribution of the non-trivial zeros of the Riemann zeta function. It closes the series.

Paper 1 [1] decomposed the explicit formula into a curvature structure and isolated the prime-local contributions that carry it.

Paper 2 [2] constructed the renormalised weights c_p^{ren} and connected the local curvature to the Weil functional.

Paper 3 [3] introduced the finite-cutoff Hilbert-space model, the prime space H_{str} , the ordinate space H_{null} , and the energy-asymmetry functional η_{orig} measuring the energy defect between them.

Paper 4 [4] exhibited the dual operator and found, on its tested finite-cutoff grid, no numerical support for a Hilbert–Pólya interpretation: the tested eigenvalues are not the ordinates, and the inverse-spectrum correlation degrades with cutoff. No structural obstruction theorem was proved there.

Paper 5 [5] proved the spectral trace formula $\text{Tr} \tilde{T}(\sigma) = D_{\text{SEL}} - O(\sigma)$ and fixed the constant D_{SEL} .

Paper 6 [6] established the curvature identity $O''(\frac{1}{2}) = -2B$ at the critical line.

Paper 7 [7] analysed stationarity of O' under an explicitly named logarithmic-saving hypothesis, which asks only for a logarithmic saving in the ordinate aspect relative to a trivial bound and is not known to follow from the Riemann Hypothesis.

Paper 8 [8] proved unconditionally that the second-moment bias of the full Guinand–Weil object is negative, and controlled hypothetical off-critical zeros using the verified height of Platt and Trudgian [9].

The present paper asks the question that characterises the object rather than extending it: what does this coupling hear, and what can it provably not hear? The answer is a classification along three axes, and the barrier axis is stated first because it governs how the other two may be read.

The operator-theoretic approach to the zeros has a long history, and the present construction sits at a definite distance from its main lines. In the programme of Connes and collaborators the spectral realisation is adelic and the primes enter through truncated Euler products, that is through a prime *cutoff*; recent work in that direction [10, 11] and its independent numerical implementations [12] keep this architecture. Here the primes are instead the *index set* of the operator, and the ordinates enter as frequencies; the difference is architectural and is set out in § 7. The semiclassical picture of Berry and Keating, and the random-matrix statistics descending from Montgomery [13] and Odlyzko [14] through Bogomolny and Keating [15], Rudnick and Sarnak [16] and Katz and Sarnak [17], provides the background motivation for comparing unfolded

worlds in §6; it is neither an input nor a hypothesis tested by the registered battery.

1.2 Main Results

§2 records the construction and the two identities imported from the earlier papers, and adds two results published here for the first time. The kernel identity $T_{p,q} = \frac{1}{2}[G(\log(p/q)) - G(\log(pq))]$ shows that the companion prime-space Gram operator $T = \Phi^*\Phi$ is the damped inverse spectrum sampled at prime ratios and prime products: one function of one variable, contracted in two ways. The canonicity lemma establishes character rigidity: within the unimodular trigonometric almost periodic class with pairwise orthogonal dilates, the building block is forced to be a character. The frequency assignment $c = \sigma\gamma_k$ and the passage to the odd part sin are recorded conventions (§2).

§3 proves that, at the reference data (100, 0.05), the normalised leading spectral ratio does not converge as the cutoff grows, with a certified two-point separation $\Delta_0 \geq 0.0840112$; that the energy-asymmetry functional $\eta_{\text{orig}}(\kappa)$ diverges to $-\infty$ in mean, an exhaustive scan over prime cutoffs $\kappa \leq 2 \cdot 10^5$ finding the last positive value at $\kappa = 146521$ and a negative one at the next prime cutoff 146527; and that the ordinate count enters only through an explicit tail bound, so that the genuine limit parameters are (κ, ε) and not (κ, N) .

§4 shows that both scaling laws are invariant across the Beurling class $\mathcal{D}_\delta$: they hold over every admissible system, with the *same* limiting function, the *same* certified separation and the *same* leading profile, whose numerical band is class-independent — a within-class non-distinction, not an ordinate-side separation. What these observables provably retain of the prime input, at the level of the proved leading laws, is the density class: R , Δ_0 , F and the band are class-independent. Lower-order, system-dependent contributions are not excluded. A sideband transfer makes the response precise for a different object, a smooth generalised-prime measure with a bounded oscillating density: the laws are not destroyed, the limiting objects move by explicit replacements.

§5 contrasts the primes with three control objects, kept separate throughout: the smooth measure μ_{sm} , whose own zeta function is zero-free; its rescaling μ_{norm} , which is the one matching the prime count exactly; and the discrete equal-mass quadrature world A . A summation formula and a quadrature estimate control the main terms; the curvature does not follow, and the discrepancy against μ_{sm} and against A is certified uniformly on the reference window. No functional of the counting data with bounded-Lipschitz constant below 35.51 can factor the discrepancy against A .

§6 states the symmetry barrier and reports the pre-registered unfolded discrimination test as a measurement. §7 assembles the classification and delineates it against neighbouring programmes. §8 documents the ordinate protocol and the certificate stack; §9 leaves the open questions.

1.3 Reader Contract

This paper *proves* the kernel identity and the canonicity of the phasor kernel (§ 2), the scaling laws, stated at the reference data (100, 0.05) (§ 3), the class transfer with the same limiting objects and the sideband transfer for a smooth oscillating measure (§ 4), the main-term equality and the factorisation barrier — the latter proved from certified inputs (§ 5) — and the symmetry barrier (§ 6). These are unconditional mathematical statements.

This paper *establishes numerically* the outcome of the pre-registered discrimination test, the completeness counts supporting it, the embedding of the sideband transfer in an explicit test world, the robustness of the second moment under completion of the mode set to prime powers, and the function-side reconstruction experiment of § 7. These statements are verified on finite grids and are labelled *numerical* in their headings. A smaller group of statements is labelled *certified*: they are bounds obtained by interval arithmetic with an explicit error protocol, named where they occur.

This paper *leaves open* the questions collected in § 9: pointwise finality of the divergence, the minimal order break, the exceptional measure under intermediate conditions, the family form of the factorisation barrier, the identification design for the prime-side separation, pair statistics on the unfolded axis, and the combinatorial gap in the canonicity lemma.

Results imported from the earlier papers are marked *proved (cited)* and are not reproved here; the citation names the paper and the section in which the proof stands.

2 The Coupling Operator: Construction, Trace Identity, and Kernel

The construction is explicit and finite. Fix κ, ε, N . The matrix $\Phi(\sigma)$ has one column per prime $p \leq \kappa$ and one row per ordinate γ_k , and $\tilde{T}(\sigma) = \Phi\Phi^*$, $T(\sigma) = \Phi^*\Phi$ are the two Gram operators attached to it. Since Φ is a finite real matrix, T and $\tilde{T}$ are positive semidefinite and

$$\text{spec}(T) \setminus \{0\} = \text{spec}(\tilde{T}) \setminus \{0\};$$

nothing is postulated about either spectrum.

Theorem 2.1 (Trace formula and curvature identity; proved (cited)). *At the reference parameters, $\text{Tr } \tilde{T}(\sigma) = D_{\text{SEL}} - O(\sigma)$ with $D_{\text{SEL}} = 10.9845$ ([5], § 3), and $O''(\frac{1}{2}) = -2B$ ([6], § 6). Numerically $B = -19342.5476$, hence $O''(\frac{1}{2}) = +38685.0952$, while the first derivative is small but not zero, $O'(\frac{1}{2}) = +2.4751$, in the three-term decomposition of that derivative ([7], § 4); the ratio of the two quantifies proximity to stationarity without establishing it. The second-moment curvature of the full Guinand–Weil object is negative unconditionally at the reference cutoff ([8], §§ 5–6).*

The first new observation is that the prime-space Gram operator $T = \Phi^*\Phi$ is an arithmetic kernel evaluated at prime ratios and products. The ordinate-space companion $\tilde{T} = \Phi\Phi^*$ has the

same non-zero spectrum, but the following identity is an entrywise formula for T , not for $\tilde{T}$.

Theorem 2.2 (Kernel identity; proved). *For all primes $p, q \leq \kappa$ and all σ ,*

$$T_{p,q} = \frac{1}{2} [G(\log(p/q)) - G(\log(pq))], \quad G(u) = \sum_{k=1}^N e^{-\varepsilon^2 \gamma_k^2} \cos(\sigma \gamma_k u).$$

The identity is algebraic and exact; it holds entrywise, for every choice of ordinate set and every regularisation width.

Proof. By definition

$$T_{p,q} = \sum_k e^{-\varepsilon^2 \gamma_k^2} \sin(\sigma \gamma_k \log p) \sin(\sigma \gamma_k \log q).$$

The product formula $\sin a \sin b = \frac{1}{2} [\cos(a - b) - \cos(a + b)]$ with $a = \sigma \gamma_k \log p$, $b = \sigma \gamma_k \log q$ gives $a - b = \sigma \gamma_k \log(p/q)$ and $a + b = \sigma \gamma_k \log(pq)$. Summing over k and using the definition of G yields the statement. $\square$

Thus the single function G determines every entry of the prime-space Gram matrix through its values at $\log(p/q)$ and $\log(pq)$: the first channel measures logarithmic differences, the second logarithmic sums. One consequence is a resolution scale. Two primes whose logarithms are close are seen by G through nearly the same argument, and their columns become nearly parallel. *Numerical observation* (resolution diagnostics). At the reference parameters the Gram correlations of the twin pairs (41, 43) and (29, 31) reach 0.937 and 0.930 respectively, while distant pairs such as (47, 53) and (2, 3) give 0.134 and -0.521 . Counting pairs above the 0.9 correlation threshold — the criterion used for fusion throughout — against the sixteen active primes leaves fourteen resolvable prime directions at the reference parameters. The correlation is not monotone in the logarithmic distance, because the product term $G(\log(pq))$ modulates it.

2.1 Canonicity of the phasor kernel

The kernel $\sin(\sigma \gamma_k \log p)$ was introduced in the earlier papers as a construction. What is forced — and all that is forced — is its character part; the remaining steps are conventions, and they are recorded as such below. Write $M[f]$ for the Bohr mean of an almost periodic f .

Definition 2.3 (Bohr mean and dilate orthogonality). *For a Bohr almost periodic function $f : \mathbb{R} \rightarrow \mathbb{C}$ write*

$$M[f] := \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T f(x) dx$$

for its Bohr mean, and

$$g \sim \sum_{\lambda \in \mathcal{F}} a_\lambda e^{i\lambda x}$$

for its Bohr–Fourier series, where $\mathcal{F} = \{\lambda : a_\lambda \neq 0\}$ is the (countable) spectrum and $\sum_\lambda |a_\lambda|^2 = M[|g|^2]$ (Parseval). We call the dilate family $\{g(\alpha \cdot)\}_{\alpha > 0}$ pairwise orthogonal if

$$M_x[g(\alpha x) \overline{g(\beta x)}] = 0$$

for all $\alpha \neq \beta > 0$.

Here g is a candidate kernel, a unimodular almost periodic function — not the damped kernel G of Theorem 2.2, which is neither unimodular nor a candidate.

Lemma 2.4 (Character rigidity under unimodularity and dilate orthogonality; proved). *Let*

$$g(x) = \sum_{\lambda \in \mathcal{F}} a_\lambda e^{i\lambda x}$$

be a trigonometric polynomial ($\mathcal{F} \subset \mathbb{R}$ finite, all $a_\lambda \neq 0$) with

(H1) $|g(x)| = 1$ for all $x \in \mathbb{R}$ (pointwise unimodularity),

(H2)

$$M_x[g(\alpha x) \overline{g(\beta x)}] = 0$$

for all $\alpha \neq \beta > 0$ (dilate orthogonality).

Then $g(x) = \omega_0 e^{icx}$ with $|\omega_0| = 1$ and a uniquely determined frequency $c \in \mathbb{R} \setminus \{0\}$. Conversely, every such character satisfies (H1)–(H2); in particular e^{icx} and e^{-icx} both do, so the hypotheses determine the character only as a $\pm c$ -pair, while c is unique for each individual g .

Proof. Step 1 (unimodularity kills multi-frequency spectra). Expanding the finite sum,

$$|g(x)|^2 = \sum_{\lambda, \mu \in \mathcal{F}} a_\lambda \overline{a_\mu} e^{i(\lambda - \mu)x} \equiv 1.$$

By uniqueness of the coefficients of a trigonometric polynomial, for every $\tau \neq 0$,

$$\sum_{\substack{\lambda, \mu \in \mathcal{F} \\ \lambda - \mu = \tau}} a_\lambda \overline{a_\mu} = 0. \tag{1}$$

Suppose $\#\mathcal{F} \geq 2$ and set $\lambda^* := \max \mathcal{F}$, $\lambda_* := \min \mathcal{F}$, $\tau_0 := \lambda^* - \lambda_* > 0$. The only pair $(\lambda, \mu) \in \mathcal{F}^2$ with $\lambda - \mu = \tau_0$ is (λ^*, λ_*) , since $\lambda \leq \lambda^*$ and $\mu \geq \lambda_*$ force equality in both. Hence (1) at $\tau = \tau_0$ reads $a_{\lambda^*} \overline{a_{\lambda_*}} = 0$, contradicting $a_\lambda \neq 0$ on $\mathcal{F}$. Thus $\mathcal{F} = \{c\}$ and, by Parseval, $|a_c|^2 = M[|g|^2] = 1$, i.e. $g = \omega_0 e^{icx}$ with $|\omega_0| = 1$.

Step 2 (orthogonality excludes the constant). If $c = 0$ then $g \equiv \omega_0$ and

$$M[g(\alpha x) \overline{g(\beta x)}] = |\omega_0|^2 = 1 \neq 0$$

for every $\alpha \neq \beta$, contradicting (H2). Hence $c \neq 0$.

Converse. For $g = \omega_0 e^{icx}$ with $c \neq 0$ and $\alpha \neq \beta > 0$ one has $M[e^{ic(\alpha-\beta)x}] = 0$ because $c(\alpha - \beta) \neq 0$, and $|g| \equiv 1$ trivially. $\square$

Remark (Division of labour between the hypotheses). *Neither hypothesis is redundant, and within the trigonometric class their roles are sharply asymmetric. $g = \cos(cx)$ satisfies (H2) — the dilate resonances $c\alpha = \pm c\beta$ require $\alpha = \beta$ or $\alpha = -\beta$, both excluded — but violates (H1) and is not a character: (H2) alone is insufficient. Conversely $g(x) = e^{i \sin x}$ satisfies (H1) with infinite spectrum $\mathbb{Z}$ (Jacobi–Anger, coefficients $J_n(1)$; identity $\sum_n J_n(1)J_{n+k}(1) = \delta_{k0}$ realises (1)), yet violates (H2): for irrational β/α the only dilate resonance is $(0, 0)$, giving*

$$M[g(\alpha x)\overline{g(\beta x)}] = J_0(1)^2 \approx 0.5855 \neq 0.$$

So (H1) alone is insufficient once the spectrum is infinite. Inside the trigonometric class, Step 1 shows that (H1) alone already forces a single frequency; (H2) contributes exactly the exclusion of the constant character.

Proposition 2.5 (Resonance identities in the full Bohr class; proved). *Let g be Bohr almost periodic with Bohr–Fourier series*

$$\sum_{\lambda \in \mathcal{F}} a_\lambda e^{i\lambda x}.$$

Then for all $\alpha, \beta > 0$,

$$M_x \left[g(\alpha x) \overline{g(\beta x)} \right] = \sum_{\substack{\lambda, \mu \in \mathcal{F} \\ \alpha\lambda = \beta\mu}} a_\lambda \overline{a_\mu}. \quad (2)$$

If in addition $|g| \equiv 1$ pointwise (H1), then (1) holds for every $\tau \neq 0$, and every difference $\tau \neq 0$ realised in $\mathcal{F}$ is realised by at least two distinct pairs. If, moreover, the dilates of g are pairwise orthogonal (H2), then:

- (a) $0 \notin \mathcal{F}$: by (H2) the left-hand side of (2) vanishes for all $\alpha \neq \beta$; choosing β/α outside the countable set

$$\{\lambda/\mu : \lambda, \mu \in \mathcal{F}, \mu \neq 0\} \cup \{1\}$$

leaves $(0, 0)$ as the only resonance on the right-hand side, forcing $|a_0|^2 = 0$.

- (b) *Ratio rigidity: every ratio $r \neq 1$ realised by a same-sign pair in $\mathcal{F}$ is realised by at least two distinct pairs (otherwise (2) with $\beta/\alpha = r$, whose left-hand side vanishes by (H2), reduces to a single nonzero term).*

All sums converge absolutely (Cauchy–Schwarz on ℓ^2 coefficients).

Proof. $g(\alpha \cdot)$ is almost periodic with Bohr–Fourier series $\sum_\lambda a_\lambda e^{i(\alpha\lambda)x}$, since the Bohr mean is invariant under positive dilations. Identity (2) is the two-function Parseval identity for Bohr

(hence Besicovitch- B^2) almost periodic functions [18, 19] applied to the pair $g(\alpha \cdot), g(\beta \cdot)$. For (1): $|g|^2 \equiv 1$ gives $M[|g|^2 e^{-i\tau x}] = 0$ for $\tau \neq 0$, while the two-function Parseval identity applied to g and $e^{i\tau x} g$ evaluates the same mean as

$$\sum_{\lambda-\mu=\tau} a_\lambda \overline{a_\mu}.$$

Absolute convergence follows from Cauchy–Schwarz, the sums being inner products of shifted, resp. rescaled-index, ℓ^2 families. The difference statement is immediate from (1). For (a) and (b), hypothesis (H2) states that the left-hand side of (2) vanishes for all $\alpha \neq \beta$; the conclusions then follow as indicated in the statement. That (H2) cannot be dropped is shown by $g \equiv 1$: Bohr almost periodic and pointwise unimodular, yet $\mathcal{F} = \{0\}$. $\square$

Remark (Canonicity of the kernel $\sin(\sigma\gamma \log p)$). *Lemma 2.4 shows that, within the class of unimodular trigonometric almost periodic kernels with pairwise orthogonal dilates, the building block is forced to be the character $x \mapsto e^{icx}$ (up to constant phase and the sign of c); that much is rigid — the character part admits no alternative within the class. What follows is a choice, and it is recorded as one: of the real functions spanned by the $\pm c$ -pair, the even one does not vanish at the origin and the odd one does, so the convention that a mode contributes nothing at $\log p = 0$ selects $\sin(cx)$ up to a scalar. The substitutions $c = \sigma\gamma_k$, $x = \log p$ then produce exactly the kernel $\sin(\sigma\gamma_k \log p)$ used throughout the series. Class definitions that vary the input data therefore run against a kernel whose character part is rigid, with projection and frequency conventions declared once and held fixed.*

For infinite spectra the conclusion of Lemma 2.4 is open. The analytic ingredients are proved above — (2) for all Bohr functions, (1) under (H1), and the conclusions of Proposition 2.5 under (H1)–(H2) — so what is missing is purely combinatorial; the precise question is recorded in §9. The trigonometric class suffices here, since the phasor kernel has finite spectrum. Beyond the single citation of the two-function Parseval identity in the proof of Proposition 2.5, no almost periodic machinery is used anywhere in this series.

2.2 Completion of the mode set

Remark (Restriction to prime modes as a design choice; numerical). *The coupling operator is built from unweighted prime phasors $\sin(\sigma\gamma_k \log p)$. Completing the mode set to all prime powers $n = p^m \leq \kappa$, which gives 24 modes at $\kappa = 53$, leaves every structural conclusion intact. The Gram identity of Theorem 2.2 holds verbatim for the extended label set: its proof uses only $\sin a \sin b = \frac{1}{2}[\cos(a-b) - \cos(a+b)]$ and the logarithmic labels, never the primality of the label, so $T_{m,n} = \frac{1}{2}\{G(\log(m/n)) - G(\log(mn))\}$ for the completed modes as well — an algebraic identity, with the residual 0 at 30 digits serving as a reproduction check rather than as its ground. Further, the curvature stays sign-definite negative across $\varepsilon \in [0.04, 0.07]$; and the spectral profile is unchanged, the leading mode remaining the one associated with γ_1 , while*

$$\lambda_{\max}/\text{Tr} \quad \text{and} \quad \lambda_2/\lambda_1$$

shift by under 4%. The numerical effect of the completion on the curvature at $\varepsilon = 0.05$ is small and depends on the — genuinely underdetermined — power-mode weight: +3.8% for the weight $\Lambda(n) \log n$, +1.3% for $\Lambda(n)^2$, and +10.8% for the operator’s own second-moment weight $(\log n)^2$. The completion does not reproduce the Guinand–Weil prime side: the raw phasors carry no functional-equation factor $n^{-1/2}$, whereas the explicit-formula prime term weights each prime power by $\Lambda(n)/\sqrt{n}$ [20]. The restriction to primes is therefore a modelling convention with quantified robustness: the curvature shifts are sign-stable and lie between 1.3% and 10.8% depending on the power-mode weight; separately, the two quoted spectral ratios move by less than 4%. It is not a truncation of a canonical prime-power sum.

3 Scaling Laws of the Renormalised Spectrum

Write

$$r(x) := \frac{\lambda_{\max}(\tilde{T}(x))}{\text{Tr } \tilde{T}(x)}, \quad u := \log x,$$

for the normalised leading spectral ratio at prime cutoff x .

Theorem 3.1 (Non-convergence of the normalised leading ratio; proved, separation certified). *At $\sigma = \frac{1}{2}$ and for the reference ordinate and regularisation data $(N, \varepsilon) = (100, 0.05)$, the limit $\lim_{x \rightarrow \infty} r(x)$ does not exist. Quantitatively,*

$$\limsup_{x \rightarrow \infty} r(x) - \liminf_{x \rightarrow \infty} r(x) \geq \Delta_0, \quad \Delta_0 := 0.0840112.$$

Strategy of the proof. Four steps: a uniform elementary asymptotic for prime sums of $p^{i\theta}$ over the finite frequency set involved; entrywise convergence of $\tilde{T}(x)/\pi(x)$ to an explicit quasi-periodic limit matrix $Q(u)$; transfer to eigenvalues by Weyl’s inequality, which needs no spectral gap; and a recurrence argument producing two values of u that are approached infinitely often, at which the limiting ratio $R(u) = \lambda_{\max}(Q(u))/\text{Tr } Q(u)$ is certified to differ.

Proof. Step 1. With

$$h(\theta, u) := \Re[e^{i\theta u}/(1 + i\theta)] = (\cos \theta u + \theta \sin \theta u)/(1 + \theta^2),$$

partial summation from the prime number theorem with classical error term gives

$$\frac{1}{\pi(x)} \sum_{p \leq x} \cos(\theta \log p) = h(\theta, \log x) + O(1/\log x),$$

uniformly over the finitely many frequencies occurring below; since $|1 + i\theta| \geq 1$ there is no small denominator, so no lower bound on $|\theta|$ is needed.

Step 2. Apply this entrywise with $w_k = e^{-\varepsilon^2 \gamma_k^2/2}$ to

$$\tilde{T}_{kl}(x) = w_k w_l \sum_{p \leq x} \sin(\sigma \gamma_k \log p) \sin(\sigma \gamma_l \log p), \quad \sigma = \frac{1}{2}.$$

The product formula used in Theorem 2.2 turns each entry into a difference of two cosine sums, and yields the real symmetric limit matrix

$$Q_{kk}(u) = \frac{w_k^2}{2}(1 - h(\gamma_k, u)), \quad Q_{kl}(u) = \frac{w_k w_l}{2} \left(h\left(\frac{\gamma_k - \gamma_l}{2}, u\right) - h\left(\frac{\gamma_k + \gamma_l}{2}, u\right) \right) \quad (k \neq l),$$

with $\|\tilde{T}(x)/\pi(x) - Q(\log x)\| \leq K/\log x$ for an explicit K . The map $u \mapsto Q(u)$ is quasi-periodic: it is a continuous function along a Kronecker flow on a torus whose dimension is the number of rationally independent frequencies among the γ_k and their half-sums and half-differences.

Step 3. Since $|h(\theta, u)| \leq (1 + \theta^2)^{-1/2}$ for all u , the trace is bounded below uniformly,

$$\mathrm{Tr} Q(u) \geq \frac{W_{\varepsilon, N}}{2} (1 - (1 + \gamma_1^2)^{-1/2}) = \tau_0 = 0.6380807 \dots > 0$$

with

$$W_{\varepsilon, N} := \sum_{k=1}^N e^{-\varepsilon^2 \gamma_k^2} = \sum_k w_k^2 = 1.37306$$

at the reference data, and $\|Q(u)\| \leq \|Q(u)\|_F \leq W_{\varepsilon, N}$, so $R(u)$ is well defined, bounded by $W_{\varepsilon, N}/\tau_0 \leq 2.152$, and Lipschitz in u with an explicit constant. Weyl's inequality transfers Step 2 to $\lambda_{\max}$ without any gap assumption, giving $|r(x) - R(\log x)| \leq K'/\log x$.

Step 4. By Dirichlet's pigeonhole principle the flow returns: there are $\tau_n \rightarrow \infty$ with all relevant frequencies simultaneously close to integer multiples of 2π , hence $\|Q(u + \tau_n) - Q(u)\| \rightarrow 0$ uniformly in u . Therefore every value of R is a limit point of r . At the two explicit points $u_1 = 17.6170$ and $u_2 = 7.1815$ the certificate below gives $R(u_1) \geq 0.5169091132$ and $R(u_2) \leq 0.4328979030$, whence the stated separation. Finally r is a step function, constant between consecutive primes, so its value set over real cutoffs coincides with that over prime cutoffs, and $\limsup$ and $\liminf$ agree in the two readings. $\square$

Certification. The separation of Step 4 is certified by ball arithmetic, not by floating-point evaluation. The lower bound on $\lambda_{\max}(Q(u_1))$ is the Rayleigh quotient $\langle v, Qv \rangle / \langle v, v \rangle$ of an explicit vector, which is rigorous for any v whatever; the upper bound on $\lambda_{\max}(Q(u_2))$ is $(\mathrm{Tr} Q^m)^{1/m}$ at $m = 64$, which majorises the largest eigenvalue for even m without any positivity assumption; both traces are enclosed directly and the two ratios are formed by outward-rounded interval division. The ordinates enter as certified enclosures — Arb's isolation of the first hundred zeta zeros, whose ball midpoints are cross-checked against the normative double-precision layer — so the enclosures are themselves part of the certificate rather than a hypothesis about it, and their radii are propagated through every operation. The certificate returns

$$R(u_1) - R(u_2) \geq 0.0840112102,$$

and is produced by `cert_paper9_ratio.py` (§8).

Definition 3.2 (Averaging operator and ordinate profile). *For locally integrable ψ on $[2, \infty)$ set*

$$\langle \psi \rangle_X := \frac{1}{\log X} \int_2^X \psi(x) \frac{dx}{x}, \quad \langle \psi \rangle := \lim_{X \rightarrow \infty} \langle \psi \rangle_X$$

whenever the limit exists. This logarithmic window mean is the only averaging operator used in this paper; no other normalisation appears. The ordinate profile is

$$f(x) := \sum_{k=1}^N e^{-\varepsilon^2 \gamma_k^2} \left(\Im \frac{x^{i\gamma_k}}{\frac{1}{2} + i\gamma_k} \right)^2.$$

Lemma 3.3 (Spectrum of the ordinate profile and its leading profile; proved). Write $\frac{1}{2} + i\gamma_k = \varrho_k e^{i\phi_k}$ with $\varrho_k^2 = \frac{1}{4} + \gamma_k^2$, and set

$$b_k := e^{-\varepsilon^2 \gamma_k^2} / (2\varrho_k^2).$$

Then f is, in the variable $u = \log x$, an exponential polynomial with frequency set $\{0\} \cup \{\pm 2\gamma_k\}$,

$$f(e^u) = \langle f \rangle - \sum_{k=1}^N b_k \cos(2\gamma_k u - 2\phi_k), \quad \langle f \rangle = \sum_{k=1}^N b_k = \frac{1}{2} \sum_{k=1}^N \frac{e^{-\varepsilon^2 \gamma_k^2}}{\frac{1}{4} + \gamma_k^2} > 0,$$

and $\langle f \rangle = 0.0021613227$ at the reference parameters. The associated leading profile

$$F(U) := \sum_{\nu} \frac{c_{\nu} e^{i\nu U}}{1 + i\nu} = \langle f \rangle - \sum_{k=1}^N b_k \Re \frac{e^{i(2\gamma_k U - 2\phi_k)}}{1 + 2i\gamma_k},$$

where $f(e^u) = \sum_{\nu} c_{\nu} e^{i\nu u}$, is Bohr almost periodic with Bohr mean $\langle F \rangle = c_0 = \langle f \rangle$ and satisfies

$$|F(U) - \langle f \rangle| \leq \sum_{k=1}^N \frac{b_k}{\sqrt{1 + 4\gamma_k^2}} \leq \frac{\langle f \rangle}{\sqrt{1 + 4\gamma_1^2}},$$

$$\text{hence} \quad F(U) \geq \langle f \rangle \left(1 - \frac{1}{\sqrt{1 + 4\gamma_1^2}} \right) > 0$$

for all U : the second inequality uses only $b_k > 0$ and $\gamma_k \geq \gamma_1 > 0$, so F is bounded away from zero with no numerical input. At the reference parameters the sharper half-width evaluates to $\sum_k b_k / \sqrt{1 + 4\gamma_k^2} = 6.7448532 \cdot 10^{-5}$, giving the band $F(U) \in [0.0020938741, 0.0022287713]$, and the exact envelope above evaluates to 0.0020849161; these three decimals are finite evaluations at the reference parameters and carry no proof step [numerical].

Proof.

$$\Im \left[\frac{x^{i\gamma_k}}{\frac{1}{2} + i\gamma_k} \right] = \varrho_k^{-1} \sin(\gamma_k u - \phi_k),$$

and $\sin^2 \vartheta = \frac{1}{2}(1 - \cos 2\vartheta)$ gives the displayed expansion with b_k as stated; the frequency set is $\{0\} \cup \{\pm 2\gamma_k\}$ and the sum is finite, so no convergence question arises. The Bohr mean of $e^{i\nu U}$ vanishes for $\nu \neq 0$, whence $\langle F \rangle = c_0$, and $c_0 = \langle f \rangle$ because dividing by $1 + i\nu$ leaves the constant mode unchanged. The first bound is the triangle inequality applied to the displayed sum, using $|1 + 2i\gamma_k| = \sqrt{1 + 4\gamma_k^2}$. The second follows termwise from $\gamma_k \geq \gamma_1 > 0$ together with $b_k > 0$, and summing gives $\sum_k b_k = \langle f \rangle$; since $\gamma_1 > 0$ the factor $1 - (1 + 4\gamma_1^2)^{-1/2}$ is positive, so the envelope is strictly positive for every N and every $\varepsilon > 0$. The decimals are finite evaluations at

the reference parameters and enter no step of the argument. $\square$

Theorem 3.4 (Divergence of the energy-asymmetry functional in mean; proved). *Let $c_p^\eta = \sqrt{V_p(\frac{1}{2})}$ and let the phasor argument be $\gamma_k \log p$, as in the paper of the series in which η_{orig} was introduced. Then, with F the leading profile of Lemma 3.3,*

$$\langle \eta_{\text{orig}} \rangle_X = -\frac{4}{W_{\varepsilon, N}} F(\log X) \frac{X}{(\log X)^3} + O\left(\frac{X}{(\log X)^4}\right) \rightarrow -\infty.$$

The leading term carries a bounded oscillating profile, not a constant: the window mean is dominated by the upper endpoint of the window, so the oscillation of f survives the averaging instead of being washed out. What makes the divergence unconditional is the exact envelope of Lemma 3.3, $F \geq \langle f \rangle (1 - (1 + 4\gamma_1^2)^{-1/2}) > 0$, together with $\langle F \rangle = \langle f \rangle$.

Proof. Step 1 (structural energy). With a_p the p -th column,

$$E_{\text{str}}(x) = \sum_{p \leq x} (c_p^\eta)^2 \|a_p\|^2,$$

where

$$(c_p^\eta)^2 = \frac{4(\log p)^2 p}{(p-1)^2} = \frac{4(\log p)^2}{p} (1 + O(p^{-1})), \quad \|a_p\|^2 = \frac{W_{\varepsilon, N}}{2} - \frac{1}{2} \sum_k w_k^2 \cos(2\gamma_k \log p).$$

Mertens' theorem gives

$$\sum_{p \leq x} (\log p)^2 / p = \frac{1}{2} (\log x)^2 + O(\log x),$$

so the constant part contributes $W_{\varepsilon, N} (\log x)^2 (1 + o(1))$. For the oscillating part, write $\vartheta(t) = t + E(t)$ with $E(t) = O(te^{-c\sqrt{\log t}})$, the prime number theorem with classical error term [20]. For fixed $\tau = 2\gamma_k \neq 0$,

$$S_\tau(x) := \sum_{p \leq x} (\log p) p^{-1+i\tau} = \int_2^x t^{-1+i\tau} d\vartheta(t),$$

$$\int_2^x t^{-1+i\tau} d\vartheta(t) = \frac{x^{i\tau} - 2^{i\tau}}{i\tau} + \left[\frac{E(t)}{t^{1-i\tau}} \right]_2^x + (1 - i\tau) \int_2^x \frac{E(t)}{t^{2-i\tau}} dt = O_\tau(1),$$

the boundary term vanishing as $t \rightarrow \infty$ and the error integral being absolutely convergent because $\int_2^\infty e^{-c\sqrt{\log t}} dt/t < \infty$ (substitute $u = \log t$). A second partial summation,

$$\sum_{p \leq x} (\log p)^2 p^{-1+i\tau} = (\log x) S_\tau(x) - \int_2^x S_\tau(t) dt/t = O_\tau(\log x),$$

then bounds

$$\sum_{p \leq x} \frac{(\log p)^2}{p} \cos(2\gamma_k \log p) = O(\log x).$$

Summing the finitely many k ,

$$E_{\text{str}}(x) = W_{\varepsilon, N}(\log x)^2 + O(\log x),$$

exactly the precision Step 3 needs for its $O(x/(\log x)^3)$ quotient error.

Step 2 (spectral energy). Since $c_p^\eta = 2(\log p)p^{-1/2}(1 + O(p^{-1}))$, partial summation from $\vartheta(x) = x + O(x/(\log x)^A)$ gives, for each fixed γ ,

$$\sum_{p \leq x} (\log p) p^{-1/2+i\gamma} = \frac{x^{1/2+i\gamma}}{\frac{1}{2} + i\gamma} + O(x^{1/2}(\log x)^{-A+1}).$$

Taking imaginary parts, the k -th component of $\sum_{p \leq x} c_p^\eta a_p$ equals

$$2w_k \sqrt{x} \Im[x^{i\gamma_k} / (\frac{1}{2} + i\gamma_k)]$$

up to the same error, so

$$E_{\text{spec}}(x) = 4x f(x) + O(x(\log x)^{-A+1}).$$

Step 3 (quotient). Hence

$$\eta_{\text{orig}}(x) = 1 - \frac{E_{\text{spec}}(x)}{E_{\text{str}}(x)} = -\frac{4x f(x)}{W_{\varepsilon, N}(\log x)^2} + O\left(\frac{x}{(\log x)^3}\right).$$

Step 4 (the average is endpoint-dominated). Substituting $x = e^u$ in Definition 3.2 turns the window mean into

$$\langle \eta_{\text{orig}} \rangle_X = -\frac{4}{W_{\varepsilon, N} \log X} \int_{\log 2}^{\log X} f(e^u) \frac{e^u}{u^2} du + O\left(\frac{X}{(\log X)^4}\right),$$

an integral whose integrand grows like e^u and which is therefore governed by its upper endpoint, not by the mean of f . Expanding $f(e^u)$ by Lemma 3.3 and integrating each mode,

$$\int^U \frac{e^{(1+i\nu)u}}{u^2} du = \frac{e^{(1+i\nu)U}}{(1+i\nu)U^2} + O\left(\frac{e^U}{U^3}\right),$$

uniformly over the finitely many frequencies ν , because $|1+i\nu| \geq 1$. Summing over the modes reproduces exactly the combination defining F , so the integral equals $F(U) e^U U^{-2} + O(e^U U^{-3})$ with $U = \log X$. Dividing by $\log X$ gives the statement. Since $F \geq \langle f \rangle (1 - (1 + 4\gamma_1^2)^{-1/2}) > 0$ by Lemma 3.3, the mean diverges to $-\infty$; the constant-rate form obtained by replacing F with its Bohr mean $\langle f \rangle$ is *not* valid pointwise in X , only on average. $\square$

Numerical observation (sign change of the quotient). The theorem is a statement about means and says nothing about individual cutoffs. Evaluating $\eta_{\text{orig}}(\kappa)$ directly at every prime cutoff $\kappa \leq 2 \cdot 10^5$, with the same ordinate list and the weights above, the quotient is positive at $\kappa = 146521$, where it equals $+0.0049$, negative at the next prime cutoff 146527 , where it equals -0.0005 , and

negative at every larger prime cutoff in the search range. For orientation, $\eta_{\text{orig}}(53) = 0.69078176$ and $\eta_{\text{orig}}(1009) = 0.810955$, while $\eta_{\text{orig}}(10^6) = -41.307$. These are evaluations on a finite grid, not part of the theorem.

Theorem 3.5 (Ordinate tail and the true limit parameters; proved). *Let $T_N = \Phi_N^* \Phi_N$ on H_{str} be the loop operator built from the first N ordinates at fixed $\sigma = \frac{1}{2}$, and T_∞ the corresponding sum over all positive ordinates; assume $\text{Tr } T_N > 0$. Then*

$$\|T_\infty - T_N\| \leq \text{Tr}(T_\infty - T_N) \leq \pi(\kappa) \sum_{k>N} e^{-\varepsilon^2 \gamma_k^2}, \quad |r_\infty - r_N| \leq \frac{2\pi(\kappa)}{\text{Tr } T_N} \sum_{k>N} e^{-\varepsilon^2 \gamma_k^2},$$

and the tail is bounded by a geometric series against the Riemann–von-Mangoldt density, so $r_N \rightarrow r_\infty$: for fixed κ and $\varepsilon > 0$ the rate is governed by the Gaussian ordinate tail $e^{-\varepsilon^2 \gamma_N^2}$, up to the Riemann–von-Mangoldt factor. The limit parameters of the construction are therefore (κ, ε) ; N is a truncation, not a parameter of the limit.

Proof. Step 1 (positive semidefiniteness and trace tail). On H_{str} the difference decomposes into rank-one terms,

$$T_\infty - T_N = \sum_{k>N} e^{-\varepsilon^2 \gamma_k^2} v_k v_k^\top$$

with $(v_k)_p = \sin(\sigma \gamma_k \log p)$, all positive semidefinite; hence the operator norm is at most the trace, and $\|v_k\|^2 \leq \pi(\kappa)$ gives the first bound.

Step 2 (ratio perturbation). For the second, Weyl’s inequality gives

$$|\lambda_{\max, \infty} - \lambda_{\max, N}| \leq \|\Delta\|$$

with $\Delta = T_\infty - T_N$, and

$$|r_\infty - r_N| = \left| \frac{\lambda_\infty}{\text{Tr } T_\infty} - \frac{\lambda_N}{\text{Tr } T_N} \right| \leq \frac{\|\Delta\| \text{Tr } T_N + \lambda_N \text{Tr } \Delta}{\text{Tr } T_\infty \text{Tr } T_N} \leq \frac{2 \text{Tr } \Delta}{\text{Tr } T_N},$$

using

$$\lambda_N \leq \text{Tr } T_N \leq \text{Tr } T_\infty.$$

Step 3 (Riemann–von-Mangoldt Gaussian tail). For the tail, split $\sum_{k>N} e^{-\varepsilon^2 \gamma_k^2}$ into dyadic blocks $\gamma \in [T, 2T)$ with $T = \gamma_N$: the Riemann–von-Mangoldt density gives $O(T \log T)$ ordinates per block, each contributing at most $e^{-\varepsilon^2 T^2}$, and the resulting series in T is dominated by a geometric one. A single-term estimate would not suffice and is not used.

Step 4 (transfer to the companion Gram operator). The statement transfers to $\tilde{T}_N = \Phi_N \Phi_N^*$ without further work: the non-zero spectra of T_N and $\tilde{T}_N$ coincide (§ 2), so trace, largest eigenvalue and hence r_N agree in the two readings. $\square$

Numerical observation (double-precision threshold). The tail bound is $2.3 \cdot 10^{-23}$ at $N = 50$ and $1.8 \cdot 10^{-61}$ at $N = 100$, against an actual $|r_{100} - r_{50}| = 2.9 \cdot 10^{-24}$. The increments therefore fall

below the double-precision unit roundoff of r well before the reference truncation: the first N with $|r_N - r_{N-1}| < 5.5 \cdot 10^{-17}$ is $N = 39$. Beyond that point the dependence on N is invisible in double precision but is not zero: at 40 digits the increments are still resolvable and strictly signed, with

$$r_{51} - r_{50} = -2.10567 \cdot 10^{-24}, \quad \text{Tr}(T_{51} - T_{50}) = +6.62880 \cdot 10^{-23},$$

so exact independence of N is false and only the limit statement above is correct. These control values were computed independently at 40 digits from the normative ordinate list; the digits beyond the tenth are not quoted, because the normative list carries double precision and does not support them.

Two further facts locate the mechanism. *Numerical observation* (leading-mode localisation). The leading eigenvector is not carried by diagonal dominance: the largest diagonal entry stays bounded as the cutoff grows, while $\lambda_{\max}$ grows like $\pi(\kappa)$. The leading mode is the wave associated with γ_1 : at cutoff $\kappa \approx 10^4$, with the 650-ordinate extended list of § 8 and $\varepsilon = 0.05$, the squared overlap $|\langle v_{\max}, e_1 \rangle|^2$ against the corresponding ordinate direction is 0.94, and it remains above 0.83 on the tested cutoff ladder up to $\kappa \approx 10^7$. Whether the oscillation of R has a pointwise, as opposed to a recurrent, limiting behaviour is a diophantine question, recorded in § 9.

4 Class Invariance over Beurling Systems

Nothing in § 3 used the multiplicative structure of the integers. This section replaces the primes by a Beurling generalised prime system and shows that the same proofs run — carrying over the limiting function R , the certified separation Δ_0 , the leading profile F and its band unchanged.

Definition 4.1 (The Beurling side). *A Beurling system is a non-decreasing unbounded sequence $\mathcal{P}_{\mathfrak{B}} = \{q_1 \leq q_2 \leq \dots\}$ with $q_1 > 1$, with counting function $\pi_{\mathfrak{B}}(x) = \#\{j : q_j \leq x\}$ and Chebyshev function $\vartheta_{\mathfrak{B}}(x) = \sum_{q_j \leq x} \log q_j$ [21, 22]. The coupling data are transported verbatim: the map*

$$\Phi_{\mathfrak{B}}(\sigma)_{k,q} = e^{-\varepsilon^2 \gamma_k^2 / 2} \sin(\sigma \gamma_k \log q), \quad q \in \mathcal{P}_{\mathfrak{B}}, \quad q \leq x,$$

the operator

$$\tilde{T}_{\mathfrak{B}}(x) = \Phi_{\mathfrak{B}} \Phi_{\mathfrak{B}}^*,$$

the ratio

$$r_{\mathfrak{B}}(x) = \lambda_{\max}(\tilde{T}_{\mathfrak{B}}(x)) / \text{Tr } \tilde{T}_{\mathfrak{B}}(x),$$

the weights $c_q^\eta = 2(\log q)\sqrt{q}/(q-1)$, the column vectors

$$a_q^\eta := \left(e^{-\varepsilon^2 \gamma_k^2 / 2} \sin(\gamma_k \log q) \right)_{k=1}^N,$$

the structural and spectral energies

$$E_{\text{str}}^{\mathfrak{B}}(x) = \sum_{q \leq x} (c_q^\eta)^2 \|a_q^\eta\|^2, \quad E_{\text{spec}}^{\mathfrak{B}}(x) = \left\| \sum_{q \leq x} c_q^\eta a_q^\eta \right\|^2,$$

and the quotient

$$\eta_{\mathfrak{B}} = 1 - E_{\text{spec}}^{\mathfrak{B}}/E_{\text{str}}^{\mathfrak{B}}.$$

Two conventions coexist here exactly as in the classical case: $r_{\mathfrak{B}}$ uses $\Phi_{\mathfrak{B}}(\frac{1}{2})$, while $\eta_{\mathfrak{B}}$ keeps the no- σ phasor convention of § 3, matching the import of η_{orig} . The ordinate side is not varied here: the γ_k remain the ordinates of the zeta function. For $\delta > 0$, $C > 0$ and $x_0 \geq 2$ let

$$\mathcal{D}_\delta(C, x_0) := \{ \mathfrak{B} : |\vartheta_{\mathfrak{B}}(x) - x| \leq C x (\log x)^{-2-\delta} \text{ for all } x \geq x_0 \}, \quad \mathcal{D}_\delta := \bigcup_{C, x_0} \mathcal{D}_\delta(C, x_0).$$

For the uniformity clause write

$$\mathcal{D}_\delta(C, x_0, M_0) := \{ \mathfrak{B} \in \mathcal{D}_\delta(C, x_0) : \pi_{\mathfrak{B}}(x_0) \leq M_0 \} :$$

a bound on the early data is genuinely additional information, because points accumulating just above 1 enter $\pi_{\mathfrak{B}}$ at full strength while contributing almost nothing to $\vartheta_{\mathfrak{B}}$.

Theorem 4.2 (Class transfer: same limiting objects and same profile; proved). *Keep the same reference ordinate and regularisation data as in Theorem 3.1, and vary only the prime-side Beurling system. Let $\mathfrak{B} \in \mathcal{D}_\delta$. Then*

$$r_{\mathfrak{B}}(x) = R(\log x) + O_{\mathfrak{B}}(1/\log x)$$

with the same limiting function R as in Theorem 3.1; the limit fails to exist for every such $\mathfrak{B}$; and

$$\limsup r_{\mathfrak{B}} - \liminf r_{\mathfrak{B}} \geq \Delta_0$$

with the same $\Delta_0 = 0.0840112$. Likewise $\langle \eta_{\mathfrak{B}} \rangle_X \rightarrow -\infty$ for every $\mathfrak{B} \in \mathcal{D}_\delta$ with the same leading profile F and the same exact positivity envelope of Lemma 3.3 in the rate of Theorem 3.4, class-independently. The objects R , u_1 , u_2 , Δ_0 , F and its Bohr mean $\langle f \rangle$ depend only on the ordinate side and are free of $\mathfrak{B}$; the certificate of Theorem 3.1 therefore transfers verbatim. Uniformity holds on each $\mathcal{D}_\delta(C, x_0, M_0)$ with the implied constants depending only on (C, x_0, δ, M_0) ; over the full class $\mathcal{D}_\delta$ it fails.

Proof. Step 1. The only property of the primes used in Step 1 of Theorem 3.1 is $\vartheta(x) = x + o(x/\log x)$, which $\mathcal{D}_\delta$ supplies by hypothesis. Split the sum at x_0 : the early points contribute at most M_0 to the sum directly and at most M_0 again through the boundary term of the partial summation at x_0 — together at most $2M_0/\pi_{\mathfrak{B}}(x)$ after normalisation — and $\pi_{\mathfrak{B}}(x) \geq (1 + o(1))x/\log x$ follows from the ϑ -condition itself, so the early-data term is absorbed in the

error. For $q > x_0$ partial summation against the ϑ -bound reproduces

$$\frac{1}{\pi_{\mathfrak{B}}(x)} \sum_{q \leq x} \cos(\theta \log q) = h(\theta, \log x) + O(1/\log x)$$

with the same h and the implied constant depending only on (C, x_0, δ, M_0) . Steps 2–4 use only this asymptotic and the fixed ordinate data, so they carry over with the same Q , the same R , and the same certified points.

Step 2. For the quotient, the Beurling form of Mertens' theorem

$$\sum_{q \leq x} (\log q)^2 / q = \frac{1}{2} (\log x)^2 + O_{\mathfrak{B}}(1)$$

follows from Definition 4.1 by partial summation, the error integral $\int^{\infty} (\log t)^{-2-\delta} dt/t$ converging precisely because $\delta > 0$; and the spectral side is again $4xf(x)$ up to a smaller error, f being built from the ordinates alone. The constants $W_{\varepsilon, N}$ and the whole profile F , band included, are therefore untouched.

Step 3 (uniformity). Within $\mathcal{D}_{\delta}(C, x_0, M_0)$ every implied constant above is a function of (C, x_0, δ, M_0) only. Over the full class $\mathcal{D}_{\delta}$, uniformity fails, and this is a separate statement about the union, established by finite deletion: deleting finitely many elements from $\mathcal{P}_{\mathfrak{B}}$ keeps $\mathfrak{B}$ in $\mathcal{D}_{\delta}$ but moves x_0 arbitrarily far out, so no threshold can be chosen uniformly. $\square$

Remark (Finite insertion near 1). *Fix (C, x_0, δ) and a base system in $\mathcal{D}_{\delta}(C, x_0, M_0)$ with fixed positive uniform slack: there exists $C' < C$ with*

$$|\vartheta_{\mathfrak{B}}(x) - x| \leq C' x (\log x)^{-2-\delta}$$

for all $x \geq x_0$. Set $m_0 := \inf_{x \geq x_0} x (\log x)^{-2-\delta} > 0$ and choose $0 < \varepsilon_0 \leq (C - C') m_0$. Adjoining K generalized primes at $q_K = \exp(\varepsilon_0/K) < x_0$ costs only ε_0 in ϑ , independently of K , while raising the count at x_0 by K ; with $b(x) := x (\log x)^{-2-\delta}$ the perturbed system stays in the class, since

$$|E_{\text{new}}(x)| \leq C' b(x) + \varepsilon_0 \leq C b(x).$$

The normalized cosine averages used in Step 1 of the proof are then dominated by the inserted points, so no equidistribution-based error control depending only on (C, x_0, δ) can be uniform over $\mathcal{D}_{\delta}(C, x_0)$. The observables themselves are far less sensitive: the inserted points shift $r(x)$ by $O(1/K)$ and the windowed η -mean by an amount bounded in terms of the reference data alone, uniformly in K , within the error budget of Theorem 4.2. The construction therefore obstructs any proof route via uniform equidistribution control; it does not decide whether the conclusions of Theorem 4.2 themselves persist on $\mathcal{D}_{\delta}(C, x_0)$.

The exponent marks the majorant threshold of the present argument, and nothing stronger is

claimed. Writing $E = \vartheta_{\mathfrak{B}}(t) - t$ and summing by parts,

$$\sum_{q \leq x} \frac{(\log q)^2}{q} = \frac{1}{2}(\log x)^2 + \left[\frac{E(t) \log t}{t} \right]_2^x + \int_2^x E(t) \frac{\log t - 1}{t^2} dt,$$

the boundary term is $O((\log x)^{-1-\delta})$ and the integral is majorised by $\int^x dt/(t(\log t)^{1+\delta})$. For $\delta > 0$ this majorant converges and the structural energy retains a bounded remainder; at $\delta = 0$ it grows like $\log \log x$; for $\delta < 0$ it is $O((\log x)^{-\delta})$. These are upper bounds delivered by this argument, not matching lower bounds: whether some system at $\delta \leq 0$ nevertheless keeps a bounded remainder is part of the boundary question of Open Problem 9.2. The direction of the classical implication should be stated carefully: a sharp Mertens estimate implies the prime number theorem always, whereas the prime number theorem does not imply a sharp Mertens estimate [23]; counterexamples exist.

The next statement concerns a different object, and the difference matters. It is not a Beurling system: it is a *smooth generalised-prime measure*, a continuous density with a bounded oscillation, and its role here is to show how the constants respond to a perturbation of the density that is not a change of class.

Definition 4.3 (Smooth generalised-prime measure and the conjugate profile). *For $0 \leq a < 1$ and $\omega > 0$ let*

$$d\vartheta_{a,\omega}(t) = (1 + a \sin(\omega \log t)) dt,$$

and write

$$\tilde{h}(\varphi, u) := \Im \frac{e^{i\varphi u}}{1 + i\varphi} = \frac{\sin \varphi u - \varphi \cos \varphi u}{1 + \varphi^2}$$

for the conjugate of the density profile h of Theorem 3.1. The sine convention is the one in which the transfer below takes its stated form; with a cosine density the same computation gives $h(\theta + \omega, u) + h(\theta - \omega, u)$ in place of the difference of conjugates, and the mass factor changes accordingly. The two conventions are not interchangeable term by term.

Theorem 4.4 (Sideband transfer of the leading representation; proved (smooth family), numerical in the discrete embedding). *Assume the non-resonance condition: the frequencies γ_k and $\gamma_k \pm \omega$, $k \leq N$, are all non-zero and pairwise distinct in absolute value. Then for $d\vartheta_{a,\omega}$ the leading representation transfers with explicit replacements:*

$$h(\theta, u) \longmapsto \hat{h}(\theta, u) := h(\theta, u) + \frac{a}{2} [\tilde{h}(\theta + \omega, u) - \tilde{h}(\theta - \omega, u)],$$

the limit matrix being formed from $\hat{h}$ by the element formulas of Theorem 3.1 with $\hat{h}(0, u) = 1 + a\tilde{h}(\omega, u)$ on the diagonal, and the Bohr mean of the leading profile shifting to

$$\langle f_{\text{osc}} \rangle = \langle f \rangle + \frac{a^2}{8} \sum_{k=1}^N e^{-\varepsilon^2 \gamma_k^2} \left[\frac{1}{\frac{1}{4} + (\gamma_k + \omega)^2} + \frac{1}{\frac{1}{4} + (\gamma_k - \omega)^2} \right].$$

A bounded relative oscillation cannot change the order of the spectral energy, by linearity in

the measure; it can only move these objects. No general assertion is made that the limiting ratio formed from $\hat{h}$ is non-constant, or that $\inf_U F_{\text{osc}}(U) > 0$, for all admissible (a, ω) ; for the reference pair $(a, \omega) = (0.3, 2\pi/3)$ the numerical embedding stands as stated (0.25%).

Proof. Count-side transfer. Insert $d\vartheta_{a,\omega}$ into the normalised counting argument of Theorem 3.1, Step 1. Writing

$$\sin(\omega \log t) = (t^{i\omega} - t^{-i\omega})/2i,$$

introduce the count measure

$$d\Pi_{a,\omega}(t) := \frac{d\vartheta_{a,\omega}(t)}{\log t} = \frac{1 + a \sin(\omega \log t)}{\log t} dt,$$

so that

$$\int_2^x t^{i\theta} d\Pi_{a,\omega}(t) = \frac{x^{1+i\theta}}{(1+i\theta)\log x} + \frac{a}{2i} \left[\frac{x^{1+i(\theta+\omega)}}{(1+i(\theta+\omega))\log x} - \frac{x^{1+i(\theta-\omega)}}{(1+i(\theta-\omega))\log x} \right] + O\left(\frac{x}{\log^2 x}\right).$$

After multiplication by $\log x/x$ and taking real parts this is exactly the stated $\hat{h}$ in the reference normalization; if the perturbed mass itself is used for normalization, divide by the corresponding $\theta = 0$ term.

Spectral-energy transfer. Insert $d\vartheta_{a,\omega}$ into the partial summation of Theorem 3.4, Step 2:

$$\int_2^x t^{-1/2+i\gamma} d\vartheta_{a,\omega}(t) = \frac{x^{1/2+i\gamma}}{\frac{1}{2}+i\gamma} + \frac{a}{2i} \left[\frac{x^{1/2+i(\gamma+\omega)}}{\frac{1}{2}+i(\gamma+\omega)} - \frac{x^{1/2+i(\gamma-\omega)}}{\frac{1}{2}+i(\gamma-\omega)} \right] + O(1).$$

Under the theorem's non-resonance hypothesis the Bohr cross terms vanish, giving the printed $a^2/8$ shift: the two sidebands contribute

$$\frac{a^2}{8} \left[\left| \frac{1}{2} + i(\gamma_k + \omega) \right|^{-2} + \left| \frac{1}{2} + i(\gamma_k - \omega) \right|^{-2} \right].$$

Since the perturbation is a bounded multiplicative factor of the measure, the leading order x of the spectral energy and the order $(\log x)^2$ of the structural energy are unchanged. $\square$

Remark (Exact resonance). At $\omega = \gamma_j$ the sideband of the j -th mode lands at frequency zero and becomes a constant, which the Bohr mean no longer halves: the mean acquires the additional term

$$\frac{a^2}{2} \sum_{k: \gamma_k = \omega} e^{-\varepsilon^2 \gamma_k^2}$$

beyond the displayed formula — twice the generic contribution of that sideband. The registered probe $(a, \omega) = (0.3, 2\pi/3)$ satisfies the non-resonance condition, so all figures quoted below are unaffected.

Remark (Which quantity is being shifted). The displayed shift of the mean concerns the unnormalised reading. If one normalises by the perturbed total mass, the same computation carries a factor $(1 + a\tilde{h}(\omega, u))^{-1}$, and the shift is correspondingly smaller; the two readings differ at order

a^2 and must not be mixed. The a^2 statement above is a statement about the Bohr-mean part of the profile; the window behaviour runs through the perturbed profile F_{osc} and is not obtained by substituting a constant. All figures quoted here are unnormalised.

Numerical evidence (embedding). For the probe $(a, \omega) = (0.3, 2\pi/3)$ the formula predicts $\langle f_{\text{osc}} \rangle / \langle f \rangle = 1.04749$. An explicitly constructed discrete test world realising that density profile gives 1.0449, a deviation of 0.25%, and the predicted limit matrix matches the measured leading ratio of that world to $7 \cdot 10^{-4}$. The closed formula is proved for the smooth family; the agreement of a discrete realisation with it is numerical, and is labelled as such.

What the two scaling laws provably see of the prime side, at leading order, is the density class; the remainder terms may retain system-dependent information. At that leading order the limiting function R , the separation Δ_0 and the profile F take the same values across $\mathcal{D}_\delta$. The statement is about these observables and is not a claim that no observable whatever could distinguish two systems within the class. Two boundary facts complete the picture. Degrading the error exponent to $x(\log x)^{-\alpha}$ preserves the laws for *every* $\alpha > 0$, with degraded rates. Indeed, writing

$$E(t) = \vartheta_{\mathfrak{B}}(t) - t = O(t/(\log t)^\alpha),$$

direct Stieltjes partial summation against E gives

$$\int_2^x t^{-1/2+i\gamma} dE(t) = O_{\alpha,\gamma}\left(\frac{\sqrt{x}}{(\log x)^\alpha}\right),$$

the boundary term and the integral against $|E(t)| t^{-3/2}$ both being of this order (split the integral at $\sqrt{x}$), so the spectral energy keeps its leading term:

$$E_{\text{spec}}^{\mathfrak{B}}(x) = 4x f(x) + O_{\alpha,N,\varepsilon}(x/(\log x)^\alpha)$$

— small against x for every $\alpha > 0$, where the substitution $A = \alpha$ in the proof of Theorem 3.4 would require $\alpha > 1$. On the structural side the remainder follows the δ -case distinction recorded above under $\alpha = 2 + \delta$: bounded for $\alpha > 2$, $\log \log x$ at $\alpha = 2$, $O((\log x)^{2-\alpha})$ below — rates degrade, the leading order never does. And it is not settled here whether the window bound admits an exceptional-set control under $\mathcal{D}_\delta$ at all: the tail available to the present argument is polynomial rather than summable, which would cap any such control at density tending to zero, and the question is recorded in §9. Whether a genuine order deviation destroys the laws is open (§9).

5 The Certified Factorisation Barrier

Section 4 shows that, at the level of the proved leading laws, variation within $\mathcal{D}_\delta$ leaves R , Δ_0 , F , and the stated band unchanged; lower-order discrimination and discrimination by other observables are not excluded. This section asks what the stated curvature observable can distinguish against control objects outside that within-class comparison, and certifies the margin.

Three control objects appear and are kept strictly apart.

Definition 5.1 (The three control objects). *Let μ_{sm}^∞ be the smooth measure $dt/\log t$ on $[2, \infty)$, and let μ_{sm} be its restriction to $[2, \kappa]$, of total mass $M = \int_2^\kappa dt/\log t = 18.1846895$ at the reference cutoff; every finite comparison below runs exclusively over the restriction. Let $\mu_{\text{norm}} := (16/M)\mu_{\text{sm}}$ be its rescaling to the mass of the prime counting measure, so that μ_{norm} — and only μ_{norm} — matches the prime count exactly. Let A be the discrete quadrature world — from here on, A denotes exclusively the discrete quadrature world; the Beurling system index in § 4 is $\mathfrak{B}$ — given by the 16 nodes $a_1 < \dots < a_{16}$ in $[2, \kappa]$ determined by*

$$\int_2^{a_j} \frac{dt}{\log t} = \left(j - \frac{1}{2}\right) \frac{M}{16}, \quad j = 1, \dots, 16,$$

that is the midpoints of the 16 cells of equal μ_{sm} -mass $M/16$, carrying unit weights. Write $E_\pi(t) := \#\{p \leq t\} - \mu_{\text{sm}}([2, t])$ for the resulting counting error of π against the smooth measure. Curvature sums attached to these objects are written B_{sm} , $B[\mu_{\text{norm}}]$ and $B[A]$; the prime curvature is $B[P]$.

Remark (The smooth measure carries no zeros). *The zeta function attached to the infinite measure μ_{sm}^∞ is*

$$\zeta_{\text{sm}}(s) = \exp(E_1((s-1)\log 2))$$

for $\Re s > 1$, continued analytically to the slit plane; an exponential never vanishes, so ζ_{sm} is zero-free, and its singular behaviour at $s = 1$ is the one the density $dt/\log t$ requires. The transform of the finite restriction μ_{sm} is an entire function and carries no singularity claim; the finite comparisons of this section use only the restriction. The control side therefore carries neither an Euler product nor zeros of its own: the contrast against the primes is pure cross-coupling.

Theorem 5.2 (Summation formula for the smooth measure; proved, finite ordering certified). *For $\varphi \in C^1[2, \kappa]$,*

$$\sum_{p \leq \kappa} \varphi(p) - \int \varphi d\mu_{\text{sm}} = (16 - M) \varphi(\kappa) - \int_2^\kappa E_\pi(t) \varphi'(t) dt,$$

with, at the reference cutoff, $|16 - M| = 2.184690$ and

$$\sup_{t \leq \kappa} |E_\pi(t)| = M - 15 = 3.1846895084\dots \approx 3.184690,$$

the supremum being attained as $t \uparrow \kappa$, that is immediately before the jump at the cutoff prime, where E_π equals $15 - M$; at $t = \kappa$ itself it has already returned to $16 - M$. The identity and the reduction to finitely many candidates are proved; the identification of the maximiser among them is certified in ball arithmetic by `certify_smooth_controls.py` (v2.0), which encloses every candidate and verifies the chain $\text{lower } |E_\pi(c_j)| > \text{upper } |E_\pi(c_{j+1})|$ along the whole ranking, so that the complete finite ordering of the candidates — not only its maximiser — is certified.

Proof. Both sides are integrals against $d(\#\{p \leq t\} - \mu_{\text{sm}}) = dE_\pi$; integrating by parts on $[2, \kappa]$

and using $E_\pi(2^-) = 0$ gives the identity. The two constants are evaluations of M and of E_π ; for the second, E_π has unit upward jumps at the primes and decreases continuously in between, so on $[2, \kappa]$ the extrema of $|E_\pi|$ occur among exactly 31 one-sided values: the sixteen right values $|E_\pi(p^+)|$ and the fifteen left limits $|E_\pi(p^-)|$, $p = 3, \dots, \kappa$ — monotonicity alone decides no more than that. Each candidate is an evaluation of the same li-difference and is enclosed as a ball; the directed separation lower $|E_\pi(\kappa^-)| > \text{upper } |E_\pi(c)|$ then holds along the whole ranking, so the finite ordering is certified as a total order: the maximiser is κ^- with margin ≥ 0.1201 over the second candidate 41^- , which in turn exceeds the third, 37^- , by ≥ 0.0920 ; the right endpoint contributes only $|16 - M| = 2.184690$. $\square$

Theorem 5.3 (Quadrature estimate for the discrete world; proved). *The position-side curvature kernel is*

$$K_\varepsilon(t) := (\log t)^2 \sum_{k=1}^N e^{-\varepsilon^2 \gamma_k^2} \gamma_k^2 \cos(\gamma_k \log t), \quad t \in [2, \kappa], \quad B[\nu] := \int K_\varepsilon d\nu,$$

so that $B[P] = \sum_{p \leq \kappa} K_\varepsilon(p) = B$ and

$$B_{\text{sm}} = \int_2^\kappa K_\varepsilon(t) \frac{dt}{\log t}.$$

With this kernel, so that $B[\nu] = \int K_\varepsilon d\nu$ for any measure ν on $[2, \kappa]$, and $y(t) = \int_2^t ds / \log s$ the logarithmic-integral coordinate,

$$B[A] = \frac{16}{M} B_{\text{sm}} + E, \quad |E| \leq \frac{16}{M} \cdot \frac{M^3}{6144} \sup_{y \in [0, M]} \left| \frac{d^2}{dy^2} (K_\varepsilon \circ t)(y) \right|,$$

$$\frac{16}{M} \cdot \frac{M^3}{6144} = \frac{M^2}{384} \leq 0.861154,$$

the mass wedge $M/16 = 1.136543$ being an exact normalisation factor and not an error term. Equivalently, before the mass rescaling,

$$|B_{\text{sm}} - (M/16)B[A]| \leq (M^3/6144) \sup |d^2(K_\varepsilon \circ t)/dy^2| \leq 0.978739 \sup |\cdot|$$

: the two constants are one bound in two normalisations, and the derivative is taken in y , the variable the rule integrates, not in t .

Proof. Substitute the logarithmic-integral coordinate $y(t) = \int_2^t ds / \log s$, which maps $[2, \kappa]$ onto $[0, M]$ and carries μ_{sm} to Lebesgue measure. In that coordinate the nodes are the midpoints $y_j = (j - \frac{1}{2})M/16$ of 16 equal panels, so A is exactly the midpoint rule for $\int_0^M K_\varepsilon(t(y)) dy$. The midpoint rule on n panels of total length L has error at most $L^3/(24n^2) \sup |\partial_y^2(K_\varepsilon \circ t)|$, which with $L = M$ and $n = 16$ is $M^3/(24 \cdot 256)$ times that supremum; multiplying by the mass factor $16/M$ gives the stated constant. The second derivative is the one taken in y , not in t : it is the composed kernel that the rule integrates. $\square$

The bound of Theorem 5.3 is not sharp for oscillatory integrands. It assumes worst-case phases

and uses no cancellation, which for the curvature kernel costs about five orders of magnitude at small ε : sixteen nodes cannot resolve frequencies $\gamma \gtrsim 1/\varepsilon$. This is declared, not hidden, and it is the reason the discrimination below is certified directly rather than deduced from the quadrature estimate.

Theorem 5.4 (Certified discrimination in curvature; certified). *Uniformly for $\varepsilon \in [0.04, 0.07]$,*

$$|B[P](\varepsilon) - B_{\text{sm}}(\varepsilon)| \geq 5343.90, \quad |B[P](\varepsilon) - B[A](\varepsilon)| \geq 5416.28.$$

At the reference width the certified ball values are $B[P](0.05) = -19342.5476062333843$, $B_{\text{sm}}(0.05) = -241.150637407723$ and $B[A](0.05) = -458.627462650173$, so at the reference width the smooth measure reproduces $\approx 1.247\%$ of the prime curvature. We write $\Delta_{\text{disc}} = 5416.28$ for the margin of the discrete comparison, which is the one entering Theorem 5.5; the comparison against μ_{sm} has the smaller margin 5343.90 and is quoted for orientation only.

Certification. The quoted margins are *downward-rounded certified lower bounds*: ball arithmetic in Arb at 192 bits of precision [24], with the ε -dependence separated exactly, the smooth-measure integrals evaluated through a closed antiderivative rather than quadrature, a covering of the window by 800 subintervals with exact rational endpoints, and explicit bounds on the ε -derivative. The ordinate inputs are certified by Arb zeta-zero isolation under exact double-precision cross-check against the normative layer, with 100 Hardy- Z sign-change enclosures of radius 10^{-20} retained as a redundant second check. The artefact is named in § 8.

Theorem 5.5 (Factorisation barrier: bounded-Lipschitz resolution lower bound; proved from certified inputs). *Let*

$$\mu_P = \sum_{p \leq \kappa} (\log p)^2 \delta_p, \quad \mu_A = \sum_{j \leq 16} (\log a_j)^2 \delta_{a_j}$$

be the $(\log)^2$ -weighted counting measures on $[2, \kappa]$, the weighting being the one the curvature itself carries; their masses are 141.6150 and 139.0471. Write

$$\hat{K}_\varepsilon(t) := K_\varepsilon(t)/(\log t)^2$$

for the reduced kernel, so that $B[P] = \int \hat{K}_\varepsilon d\mu_P$ and $B[A] = \int \hat{K}_\varepsilon d\mu_A$: the $(\log)^2$ weight sits in the measures here and in the kernel of Theorem 5.3, never in both. A factoring functional is a map Ψ on finite signed measures on $[2, \kappa]$; equip that space with the flat bounded-Lipschitz distance

$$d_{\text{BL}}(\mu, \nu) := \sup \left\{ \int \varphi d(\mu - \nu) : \|\varphi\|_\infty \leq 1 \text{ and } \text{Lip}(\varphi) \leq 1 \right\},$$

the flat convention, under which the Dudley sum variant $\|\varphi\|_\infty + \text{Lip}(\varphi) \leq 1$ gives a smaller value, so the conclusion holds a fortiori. Then, with $d_{\text{BL}}(\mu_P, \mu_A)$ as certified in Lemma 5.6 below, no functional of the counting data with bounded-Lipschitz constant

$$L < \frac{\Delta_{\text{disc}}}{d_{\text{BL}}(\mu_P, \mu_A)}$$

can factor the discrimination of Theorem 5.4.

For this single pair of measures the bound is exactly the pairwise Lipschitz quotient; no stronger obstruction to factorisation is claimed.

The distance itself is a certified input, stated separately so that the theorem’s status is exact.

Lemma 5.6 (Bounded-Lipschitz distance of the weighted measures; certified). $d_{\text{BL}}(\mu_P, \mu_A) = 152.529455\dots$, *rigorously two-sided*.

Certification. The value is obtained as a linear program in the Kantorovich–Rubinstein form, solved to optimality and certified in two directions: primal feasibility of the returned φ is verified in ball arithmetic, and weak duality with residual absorption ($\|r\|_1 = 1.8 \cdot 10^{-14}$) bounds the optimum from the other side.

Proof of Theorem 5.5. If $B[P] = \Psi(\mu_P)$ and $B[A] = \Psi(\mu_A)$ for a functional Ψ with bounded-Lipschitz constant L , then

$$|B[P] - B[A]| = |\Psi(\mu_P) - \Psi(\mu_A)| \leq L d_{\text{BL}}(\mu_P, \mu_A).$$

Combining with the lower bound of Theorem 5.4 and the certified value of Lemma 5.6 gives $L \geq \Delta_{\text{disc}}/d_{\text{BL}}$. $\square$

The threshold depends on which margin is inserted, and the three figures in circulation are all the same statement at different strengths. With the margin declared in advance of the computation, 5000, the threshold is $5000/152.529455 = 32.78$; with the margin actually achieved uniformly on the window, $\Delta_{\text{disc}} = 5416.28$, it is 35.51; and pointwise, using the realised difference at a single width instead of the uniform bound, it is 123.8 at $\varepsilon = 0.05$ and 283.8 at $\varepsilon = 0.04$. The sharp uniform value 35.51 is the one quoted in the abstract.

The honest reading of Theorem 5.5 is a resolution lower bound, not an impossibility statement. The curvature is itself bounded-Lipschitz, with kernel sup-norm at most 664 at the reference width; the theorem says that any factoring functional must have a constant at least in the tens, that is, must resolve the counting data far more finely than the density does. The statement covers one pair of worlds; the family form is open, and functionals outside the bounded-Lipschitz class are untouched.

The programmatic consequence is the one that matters for §6, and two levels must be kept apart before it is drawn: the ordinate channel is external and identical in both comparisons, while the intrinsic zeta function of the control side is a separate object — it is the former that carries the coupling. On that external channel both objects compared here have the same zero side, and the intrinsic zeta function of the smooth measure has no zeros at all, so the entire discrimination is carried by the prime side. What is certified is precise and should be stated precisely: the curvature separates from the smooth measure with uniform margin ≥ 5343.90 and from the discrete world with uniform margin ≥ 5416.28 , and for the discrete pair — the pair the bounded-Lipschitz distance is attached to — no functional of the counting data with

constant below the stated threshold factors that separation. That identifies the prime-side weighted point configuration, relative to the stated controls, as audible; it does not by itself identify multiplicativity as the cause, because the controls vary the point geometry and the multiplicative structure together.

6 Limits of Audibility

The barrier comes first.

For the curvature sums the relevant carrier is the aggregated curvature test kernel

$$\mathcal{K}_{\kappa,\varepsilon}(z) := \sum_{p \leq \kappa} (\log p)^2 z^2 e^{-\varepsilon^2 z^2} \cos(z \log p),$$

a sum over *primes* in the *ordinate* variable — distinct from G , which sums over ordinates in the ratio variable; the two must not be interchanged. That $\mathcal{K}_{\kappa,\varepsilon}$, and not G , carries the continued curvature sums is an identity, not a reading: at the reference data,

$$\sum_{k=1}^N \mathcal{K}_{\kappa,\varepsilon}(\gamma_k) = B = -19342.5476,$$

the per-ordinate decomposition of the curvature anchor of the series.

Theorem 6.1 (Functional-equation symmetry barrier; proved). *Let a zero be written $\rho = \beta + i\gamma$ and let $\delta = \beta - \frac{1}{2}$ denote its distance from the critical line. The functional equation $\xi(s) = \xi(1-s)$, together with reality of ξ on the real axis, places zeros in orbits $\{\frac{1}{2} \pm \delta \pm i\gamma\}$. Every symmetric functional of such an orbit is an even function of δ . Evenness holds without any regularity hypothesis, and it already implies that no symmetric functional of the zero data — in particular none of the observables of this paper, and none of the curvature sums of the earlier papers — can determine the sign of δ . If such a functional is in addition differentiable at $\delta = 0$, its first derivative there vanishes; if it is analytic, the first possibly non-zero Taylor coefficient is of even order. For the curvature sums of this series, whose δ -dependence enters through the even orbit average*

$$\frac{1}{2}(\mathcal{K}_{\kappa,\varepsilon}(\gamma_k + i\delta) + \mathcal{K}_{\kappa,\varepsilon}(\gamma_k - i\delta)),$$

that first term is the quadratic one, unless the aggregate quadratic coefficient happens to cancel. The class of observables studied here is therefore blind to the orientation of the displacement — unconditionally. When such an observable is differentiable in δ at 0, its first derivative there vanishes; when it is real-analytic, its expansion contains only even powers, and the response to the magnitude begins at even order.

Proof. The orbit is invariant under $\delta \mapsto -\delta$ by the functional equation combined with conjugation. A symmetric functional of an invariant set is invariant under any permutation of it, hence takes the same value at δ and $-\delta$; an even function of δ has vanishing first derivative at

$\delta = 0$. For the curvature sums the mechanism is explicit: each ordinate enters the continued sums through the even orbit average

$$\frac{1}{2}(\mathcal{K}_{\kappa,\varepsilon}(\gamma_k + i\delta) + \mathcal{K}_{\kappa,\varepsilon}(\gamma_k - i\delta)) = \mathcal{K}_{\kappa,\varepsilon}(\gamma_k) - \frac{\delta^2}{2} \mathcal{K}_{\kappa,\varepsilon}''(\gamma_k) + O(\delta^4),$$

so the first-order term vanishes identically and the first possibly non-zero contribution is the aggregate quadratic one. $\square$

This is a limit, not a lament, and it is stated before the measurement so that the measurement cannot be misread. The statement is precise about what fails: not that the observables carry no trace of a displacement whatever, but that they cannot resolve its direction, and that, under differentiability, the first-order response to its magnitude vanishes — for the real-analytic curvature sums of this series, the response begins at the quadratic term, unless the aggregate quadratic coefficient happens to cancel. What the theorem bars is any inference to the *orientation* of the displacement. The even-order response can depend on $|\beta - \frac{1}{2}|$, so an inference to the *magnitude* is not excluded by the theorem; whether the magnitude is in fact reconstructible from these observables is neither proved nor tested in this paper, and no such inference is made anywhere in it.

What remains open to measurement is the ordinate side: does the coupling hear anything about the ordinates beyond their density class? The following measurement was designed and registered before it was run.

Normalisation. The ordinate axis is varied while the prime side is held fixed at $p \leq \kappa$. Ordinates are unfolded against each world's own smooth counting function, $\tilde{b}_k := N_W(b_k)$, giving unit density, and the Gaussian weight is applied on the unfolded scale with $\tilde{\varepsilon} = 0.25$; robustness is checked at $\tilde{\varepsilon} = 0.10$. Only unfolded statements carry discriminating content beyond density. The phasor argument in this battery is $\tilde{b} \log p$, without the trace parameter; this differs from the convention of §§ 2 to 5 and is declared here because the observables below are sensitive to it. The convention was fixed before the computation, not chosen after it.

Design. Five worlds, each constructed independently and given explicitly: the Riemann ordinates W_ζ ; the ordinates W_{χ_4} of the Dirichlet L -function of the non-principal character modulo 4; those of the Dedekind zeta function of $\mathbb{Q}(i)$, a degree-two Euler object assembled as a union, W_{Ded} ; those of the Davenport–Heilbronn function [26], which has a functional equation but no Euler product and of which only the imaginary ordinates enter — its off-line real parts are discarded — W_{DH} ; and a synthetic density skeleton W_{syn} with eight replicates supplying the scatter scale S_i . The skeleton is the exactly unfolded sequence $\tilde{b}_k = k - \frac{1}{2}$, $k = 1, \dots, 30$, and the replicates are

$$\tilde{b}_k^{(m)} = k - \frac{1}{2} + 0.25 \sin(2\pi\{k\phi_m\}), \quad \phi_m = \{\phi^m\}, \quad m = 1, \dots, 8,$$

with ϕ the golden ratio and $\{\cdot\}$ the fractional part: an explicit deterministic recipe, with no random number generator anywhere. The recipe degenerates, and the degeneration is worth stating precisely. Since $\phi^2 = \phi + 1$, one has $\{\phi\} = \{\phi^2\}$, so the replicates $m = 1$ and $m = 2$

coincide and only seven are distinct. Because S_i below is a maximum over the replicates, a repeated replicate contributes a value already present: S_i is numerically *unchanged* by the degeneration. What is lost is one degree of freedom in the design, not scatter.

With

$$\Phi_{ij} = e^{-\tilde{\varepsilon}^2 \tilde{b}_i^2 / 2} \sin(\tilde{b}_i \log p_j),$$

$G^{\text{un}} = \Phi^\top \Phi$ on the 16-dimensional prime space and $\tilde{T} = \Phi \Phi^\top$ on the ordinate space, the five observables fixed in advance were

	Observable	Normalisation
$\mathcal{O}_1$	$\sum_j (\log p_j)^2 \sum_i e^{-\tilde{\varepsilon}^2 \tilde{b}_i^2} \tilde{b}_i^2 \cos(\tilde{b}_i \log p_j)$	raw
$\mathcal{O}_2$	$\lambda_{\max}(G^{\text{un}}) / \text{Tr } G^{\text{un}}$	dimensionless
$\mathcal{O}_3$	$\text{Tr } G^{\text{un}}$	raw
$\mathcal{O}_4$	$(\sum_{i,j} \tilde{T}_{ij} - \sum_i \tilde{T}_{ii}) / \sum_{i,j} \tilde{T}_{ij} $	fraction
$\mathcal{O}_5$	$\lambda_2(G^{\text{un}}) / \lambda_1(G^{\text{un}})$	dimensionless

Criterion. Registered in advance, with

$$D_i(W) := |\mathcal{O}_i(W) - \mathcal{O}_i(W_{\text{syn}})|, \quad S_i := \max_{m=1,\dots,8} |\mathcal{O}_i(W^{(m)}) - \mathcal{O}_i(W_{\text{syn}})|,$$

a world W separates on $\mathcal{O}_i$ if $D_i(W) > 3S_i$. The design note fixing worlds, observables, criterion and window is dated and archived with the verification code [35].

Outcome (numerical). The criterion is not met, and not narrowly. At $\tilde{\varepsilon} = 0.25$ the scatter scales are $S_1 = 640.995$, $S_2 = 0.151280$, $S_3 = 4.71715$, $S_4 = 0.0515498$, $S_5 = 0.483144$, and the ratios D_i/S_i are

	$\mathcal{O}_1$	$\mathcal{O}_2$	$\mathcal{O}_3$	$\mathcal{O}_4$	$\mathcal{O}_5$
W_ζ	0.299	0.069	0.055	0.580	0.135
W_{χ^4}	0.342	0.031	0.064	0.324	0.073
W_{Ded}	0.154	0.630	0.586	0.270	0.270
W_{DH}	0.082	0.060	0.018	0.168	0.012

The largest entry of the whole matrix is 0.630, a factor 4.8 below the threshold; not one cell reaches even one scatter unit. At $\tilde{\varepsilon} = 0.10$ the picture is unchanged, with maximum 0.799. In particular the world with a functional equation but no Euler product does not separate either: there is no non-Euler anomaly on this axis. The outcome is qualitatively compatible with broad universality heuristics for unfolded local statistics [16], but the battery does not test an n -level correlation law; its registered conclusion is only that none of the five observables separates the declared worlds at the stated threshold.

Scope. The finding holds for this battery, this window and this coupling. Pair and correlation statistics of the unfolded ordinates were not examined; any continuation would be a new experiment requiring its own pre-registration. The effective sample behind the gate is small —

$n = (\sum_k w_k)^2 / \sum_k w_k^2 \approx 7.1$ at $\tilde{\varepsilon} = 0.25$ and 17.6 at $\tilde{\varepsilon} = 0.10$, the Kish effective size of the Gaussian gate weights $w_k = e^{-\tilde{\varepsilon}^2 \tilde{b}_k^2 / 2}$ on the 30 unfolded ordinates (`ess_gate.py`) — and this was declared in advance as the weakest point. No no-go theorem is derived from this measurement, and it says nothing whatever about the Riemann Hypothesis. The term *pre-registered* is used here as a methodological import from the experimental sciences, and is meant literally: design, observables and criterion were fixed in a dated document before the computation was run.

Completeness. That the ordinate lists are complete in the window was verified by rectangle counting via the argument principle. For the two worlds whose lists had to be constructed by an independent sign search — the character world and the world with a functional equation but no Euler product — the winding number came out at exactly 30 in each case, with contour minimum modulus at least 0.27. This is a numerical verification, not an interval-arithmetic one, and is labelled accordingly. As a by-product, all zeros of the Davenport–Heilbronn function in the window lie on the line, the first off-line ordinate, at 85.70 [30], lying above the window maximum 64.04 — a statement about the known zero list, independent of the rectangle count.

7 The Audibility Classification and Discussion

The three preceding sections classify the coupling along three axes. Throughout this paper, *audible* is used in a relative sense: a feature of the data is called audible if some observable of the stated class separates the world carrying it from the stated comparison worlds. The classification below is therefore always relative to the declared observable class and the declared controls; no absolute notion of audibility is asserted.

Axis	Finding	Status
Prime-side point configuration vs. stated controls	audible, margins ≥ 5343.90 / ≥ 5416.28	certified
Prime-side Beurling within-class variation	not distinguished at the level of the proved leading laws; same R , Δ_0 , F and band; lower-order discrimination not excluded	proved
Ordinates beyond unfolded one-point density	not separated on this bat- tery	numerical
Zero-orbit orientation	not audible; first derivative zero when differentiable	proved

The order is the point. The barrier is a theorem, so the last row is not a gap in the measurement but a boundary of the object; the third row is a measurement with a declared scope; and the second row is a proved *invariance* — within the class the proved leading asymptotic data

do not distinguish: R , Δ_0 , F and the band are class-independent. The system-dependent remainder terms are not controlled sharply enough here to exclude lower-order discrimination. The first row states what is certified — the prime-side point configuration separates from the stated controls — and a multiplicative reading of that separation, that what is heard is Euler structure, is available as an *interpretation* and is marked as one here: the controls of § 5 vary point geometry and multiplicative structure together, so the certified statement does not attribute the separation to either alone. The identification experiment that would decide this is stated as Open Problem 9.6. A function-side reconstruction experiment, reported after this discussion, is directionally compatible with this reading. Read so, the first and second rows together say that this coupling is an instrument for arithmetic point structure and for counting density — the latter as the surviving class parameter — and for nothing else that we were able to test on this finite battery.

Numerical observation (function-side reconstruction; a separate experiment, not part of the § 6 battery). The experiment asks how much of the zero placement, and of the area signature between consecutive zeros, a phasor field recovers along the critical line. Truth is Hardy’s $Z(t)$ on $t \in [100, 600]$ (grid 0.04; 312 zeros of the extended list `zeros_650.csv`, cross-checked against the normative first-100 layer, which it reproduces exactly in double precision, hence 311 areas); the models are the finite cosine sums $F(t) = \sum_n 2n^{-1/2} \cos(\theta(t) - t \log n)$ over declared index sets inside the Riemann–Siegel window $n \leq \sqrt{t/2\pi}$, with two pre-declared metrics: zero recovery at tolerance 0.25 of the local mean spacing, and Spearman rank correlation of the areas. With all amplitudes fixed at $2n^{-1/2}$, the full window set recovers 98.1% of the zeros with 0.0% spurious crossings and area correlation 0.980; prime indices alone drop recovery to 34.6% with 35.4% spurious; adding the constant mode and the prime powers $\{4, 8, 9\}$ reaches 85.9%; the single semiprime term $n = 6 = 2 \cdot 3$ closes the gap back to 98.1%. After the amplitude normalisation, the degradation pattern under removal of composite modes is consistent with sensitivity to combination frequencies of already active prime modes. The comparison with the curvature side is directional only: the curvature controls of § 5 vary multiplicative structure and point geometry together, so the two experiments are directionally compatible, but no common causal feature is established. The experiment is produced by `function_side_reconstruction.py` (§ 8).

7.1 Near-coincidence fusions

Numerical observation. The resolution scale of § 2 has a sharper consequence once the mode set is completed to prime powers. Three further pairs fuse: 31 with 32 at correlation 0.977, 16 with 17 at 0.908, and 47 with 49 at 0.907. In each case a prime power lands beside a neighbouring prime — 2^5 beside 31, 2^4 beside 17, 7^2 beside 47 — and the two directions become indistinguishable to the operator. The effect is deterministic logarithmic geometry and nothing else: the kernel sees $\log n$. Across the completed 24-mode reference set, and at the reference parameters, every pair whose logarithms differ by less than roughly 0.065 was found unresolved; this is a one-sided empirical rule for that set, not a general resolution criterion, since the pair correlation runs through $G(\log mn)$ as well and is therefore not monotone in $|\log m - \log n|$ alone. This threshold is genuinely smaller than the rule of thumb 0.12 obtained by calibrating on prime

pairs alone, which is too loose for power pairs; the pair (25, 27) at logarithmic distance 0.0770 does not fuse, reaching only 0.540, and the pair (8, 9) at 0.1178 reaches 0.224. Multiplicative structure meets additive structure at these points, and the instrument cannot tell them apart.

7.2 Delineation

The following delineations are stated rather than implied.

Prime cutoff against prime indexing. The line of work of Connes and collaborators constructs self-adjoint operators from truncated Euler products, the truncation parameter controlling which primes $p \leq c$ enter the operator [10, 11, 12]. Self-adjointness is there by construction, so the spectrum is real by design; what is approximated is the sequence of ordinates, and the entire weight of the Riemann Hypothesis rests on an unproven convergence as the cutoff grows. This does not conflict with Theorem 6.1, whose subject is different: not whether some operator can be built whose spectrum is real, but whether the distance of a zero from the critical line can be *read off* a class of observables that presupposes nothing. In the language of that programme, the scaling spectral triple plays the role of the archimedean factor — the truncated completion $\xi = \Gamma_{\mathbb{R}} \prod (1 - p^{-s})^{-1}$ in operator form — and the same ingredient that produces the zeros removes, for symmetric functionals, their location. Here the primes are the index set of a Gram operator and the ordinates are frequencies; that is a different architecture, not a variant of the same one.

The nearest neighbour. Groskin’s finite Guinand–Weil dictionary [29] establishes, for the truncated Weil quadratic form, an exact finite correspondence in which the value of the truncated form on a Galerkin coefficient vector equals a sum of an explicitly constructed band-limited test function over the non-trivial zeros. This is thematically adjacent to the trace identity of § 2: both are exact finite bridges between an operator quantity and a sum over zeros. Architecturally the two are separate. That construction lives on the test-function side, in a frequency-band basis, and its object is Weil positivity; there is no prime–ordinate Gram coupling in it, no variation of the arithmetic world, and no question of audibility. Neither result implies the other, and neither is a special case of the other.

Screw functions. Suzuki’s unification of the Weil quadratic form [31] treats the same positivity question through continuous functions rather than distributions; it is adjacent, and likewise free of any assumption of the Riemann Hypothesis. It does not index by primes.

Local Euler factors. Liu’s operator pencils [32] encode local Euler factors prime by prime and compose them as a restricted tensor product. Primes do appear there as an index family, which makes this the closest parallel track on the indexing question; but the objects are local factors, not a global coupling between primes and ordinates, and no spectral ratio, no class transfer and no audibility question arises.

Arithmetical matrices. Hilberdink and Pushnitski [34] obtain spectral asymptotics for a fixed family of arithmetical matrices, with the same constant governing both spectral branches, and indicate a connection to Beurling systems. That constant belongs to that matrix family. Theorem 4.2 establishes a different statement: the present leading objects R , Δ_0 , F , and their band

are invariant across systems in $\mathcal{D}_\delta$. It does not establish uniqueness of those objects across different density classes.

Operators over Beurling-induced spaces. Kouroupis [33] studies Hardy spaces of generalised Dirichlet series induced by Beurling primes and shows that boundedness and compactness of certain composition operators do not depend on the choice of primes. This is genuine operator theory over Beurling systems, but there is no Gram matrix, no ordinate channel, and no fingerprint in asymptotic spectral constants.

Off-line zeros. The Davenport–Heilbronn function is the standard reference for the statement that a functional equation without an Euler product does not force the zeros onto the line [26]. It is not the first documented off-line zero: that record belongs to the work of Potter and Titchmarsh on the Epstein zeta function [27, 28].

8 Methods and Verification

Ordinate protocol. Two layers are used and never mixed. The normative layer is a fixed list of the first 100 ordinates in double precision; it defines every anchor value and every tolerance quoted in this paper, and it is the reason no figure here is quoted beyond the precision that list supports. Where a certificate requires more, the ordinates are obtained as certified enclosures from Arb’s zeta-zero isolation, and the ball midpoints are subject to a mandatory cross-check against the normative layer, which they must reproduce exactly in double precision; the enclosures are then part of the certificate itself. For non-certified high-precision work the regeneration recipe remains `mpmath.zetazero` at 30 digits, under the same cross-check. No regenerated list is archived; the generating scripts are. The extended list `zeros_650.csv` is a named, non-normative auxiliary file used for exactly one numerical experiment (§ 7); it defines no anchor and no tolerance, and its first 100 entries reproduce the normative layer exactly in double precision. No further *normative* ordinate file exists.

Artefacts. Each machine-checked statement of this paper has a named artefact in the code repository [35]. The anchor and prohibition checks of the manuscript are `verify_paper9.py` (v0.30, 277 checks; the script verifies its own check count and exits with failure on any shortfall, so a partial run cannot end in success — it is a drift gate over the quoted values, recomputing the anchors marked for recomputation, and does not itself reproduce the experiments). The five-world battery of § 6 is produced by `unfolded_discrimination.py`: world construction including the Hurwitz-zeta χ_4 -line and the Davenport–Heilbronn ordinates after the functional-equation construction, the five observables with their scatter scales, and the completeness counts by the argument principle; the non- ζ ordinate lists are constructed deterministically inside the script, which therefore serves as their generation protocol; for W_ζ the script regenerates the first 31 ordinates internally with `mpmath.zetazero` at 40 digits and cross-checks them against the normative `zeros_100.csv`. The effective gate size is the one-line computation of `ess_gate.py`. Both certificate artefacts obtain their ordinates by Arb zeta-zero isolation (`acb.zeta_zeros`) under exact double-precision cross-check against the normative layer. The function-side reconstruction experiment of § 7 is produced by `function_side_reconstruction.py`: truth grid, index sets,

tolerance and both metrics are declared in the script, and the quoted recovery and correlation figures are its direct output; its ordinate input is the auxiliary file `zeros_650.csv`. The two-point separation of Theorem 3.1 is certified by `cert_paper9_ratio.py` (v1.2), which obtains the ordinates as certified enclosures from Arb’s zeta-zero isolation, cross-checks the ball midpoints against the normative double-precision layer, builds the limit matrices as ball matrices, bounds the leading eigenvalue from below by a Rayleigh quotient at u_1 and from above by a trace power at u_2 , and divides the resulting intervals outward. The margins of Theorem 5.4, the closed antiderivative replacing quadrature, the window covering and the linear program certifying d_{BL} in Theorem 5.5 are produced by `certify_smooth_controls.py` (v2.0) as downward-rounded certified lower bounds; for non-certified high-precision regeneration, `gen_zeros.py` computes the first 100 ordinates with `mpmath.zetazero` at 30 digits and cross-checks their double-precision values against the normative `zeros_100.csv`. It is not an input to either Arb certificate. Each certificate script obtains its own certified zero enclosures internally via `acb.zeta_zeros` and performs the mandatory midpoint cross-check against the normative layer.

Certificate stack. Certified statements use ball arithmetic in Arb through its Python interface, `python-flint` 0.9.0 on FLINT 3.6.0 [24]; non-certified numerics use `mpmath` 1.4.0 [25] at the precision stated in each place. The off-line control quoted from the earlier work of this series rests on the verified height $H_0 = 3 \cdot 10^{12}$ [9], an unconditional finite computation.

Reproducibility. The mechanical checks are of four kinds, and the gate distinguishes them: values marked for recomputation are recomputed from the ordinate list; textual anchors are drift anchors — their presence is enforced, their values fixed by the artefact that produced them; interval statements have dedicated certificate artefacts whose outputs are themselves checked as anchors; and the identities of Theorems 2.1 and 2.2 are checked as identities, not as approximate agreements. Plots are not reproduced here; they live in the code repository together with the scripts.

9 Open Problems

These are bequests, not a programme. Each is stated in the terms in which it would have to be answered.

Open Problem 9.1 (Pointwise finality of the divergence). *Theorem 3.4 is a statement about window means, and Theorem 3.1 about limit points.*

Given. The profile $f(e^u)$ of Lemma 3.3 and the recurrence of the Kronecker flow.

Target. A pointwise statement: does $\eta_{\text{orig}}(\kappa) \rightarrow -\infty$ without averaging?

The question splits into two tasks. First, remainder control in the small-profile regime: pointwise, Step 3 of the proof of Theorem 3.4 gives

$$\eta_{\text{orig}}(e^u) = -\frac{4e^u f(e^u)}{W_{\varepsilon, N} u^2} + O\left(\frac{e^u}{u^3}\right),$$

so once $f(e^u) \lesssim 1/u$ the leading term falls to the size of the remainder and the remainder decides the sign. Second, quantitative simultaneous approximation: under the additional, unproved hypothesis that the reference ordinates are linearly independent over $\mathbb{Q}$, Kronecker approximation gives $\inf_u f(e^u) = 0$, but supplies no rate; without that hypothesis even the infimum is not established here. What matters is therefore how often and how deeply f nearly vanishes, not that it does. The dichotomy is unresolved in both directions.

Open Problem 9.2 (Minimal order break). *Theorem 4.2 covers density classes whose remainder is $O(x/(\log x)^{2+\delta})$. Determine the minimal deviation from $\vartheta_{\mathfrak{B}}(x) \sim x$ that destroys the class laws, rather than merely degrading their rates. Whether the scaling laws hold uniformly over the sideband family of Theorem 4.4 — non-constancy of the transferred ratio and positivity of the transferred profile for all admissible (a, ω) — is likewise open. So is the status of the early-data bound: whether M_0 in the uniformity clause of Theorem 4.2 can be dispensed with at fixed (C, x_0) : whether the uniformity of the conclusions persists on $\mathcal{D}_\delta(C, x_0)$ without the early-count bound is open; any proof would have to bypass uniform equidistribution control, which finite insertion near 1 destroys (see the Remark after Theorem 4.2). It is not decided here.*

Open Problem 9.3 (Exceptional measure under intermediate conditions). *It is not established here that the window bound admits any exceptional-set control under the hypothesis of Definition 4.1; the tail available to the present argument is polynomial rather than summable, which would cap such a control, if one exists, at density tending to zero. Determine whether an exceptional-set statement holds at all under $\mathcal{D}_\delta$, of what strength, and what additional summability of the error term, if any, yields finite measure.*

Open Problem 9.4 (Family form of the factorisation barrier). *Theorem 5.5 compares one pair of worlds. Establish a barrier uniform over a family of control worlds, and determine whether the threshold constant can be pushed beyond the bounded-Lipschitz class.*

Open Problem 9.5 (Pair statistics on the unfolded axis). *The measurement of § 6 used one-point observables. Whether pair or higher correlation statistics of the unfolded ordinates separate arithmetic worlds from a density skeleton, under a criterion fixed in advance, is untested here.*

Open Problem 9.6 (Identification design for the prime-side separation). *The certified discrimination of § 5 separates the prime configuration from density-matched controls; it does not identify which feature carries the separation, because those controls vary the one-point geometry and the multiplicative structure together. Construct a family of control worlds matched to the primes in multiplicative structure while varying the one-point geometry, or matched in one-point geometry while varying multiplicativity, with margins and criterion fixed in advance — the experiment that would turn the interpretation of § 7 into a result, in either direction.*

Open Problem 9.7 (Canonicity beyond the trigonometric class). *Lemma 2.4 is proved for finite spectra. The analytic ingredients (1) and (2), and the exclusion $0 \notin \mathcal{F}$, hold in the full Bohr class by Proposition 2.5 under (H1)–(H2). What is missing is purely combinatorial: do these three conditions force $\#\mathcal{F} = 1$ for countable $\mathcal{F}$ with square-summable coefficients? The hardest model case is $\mathcal{F} \subset \mathbb{Z}$, where the conditions become fibre relations $\sum_m a_{pm} \overline{a_{qm}} = 0$. Proposition 2.5(b), under (H1)–(H2), already excludes every spectrum in which some difference or some same-sign ratio is uniquely represented.*

Non-Circularity

We document explicitly that the results of this paper do not rely on the objects they are directed toward.

No RH as input. The Riemann Hypothesis is never assumed. The ordinates enter as numerically specified inputs; no property of their distribution requiring the Riemann Hypothesis is used. Theorem 6.1 is a statement about the functional equation alone.

No GUE, Montgomery, or Hilbert–Pólya hypothesis. The distribution of the ordinates is not modelled by random matrix theory. The random-matrix literature enters only as background motivation for the comparison reported in § 6, never as an assumption and never as a hypothesis the battery tests. The coupling operator is neither postulated nor used as a Hilbert–Pólya operator; Paper 4 [4] reports a numerical diagnostic failure on its tested finite-cutoff grid. Its entries are explicit deterministic arithmetic expressions.

No inference from a discriminator to a location. The discrimination established in § 5 concerns the prime side, and two levels must be kept apart. The ordinate channel that carries the coupling is external and identical in both comparisons; separately, the intrinsic zeta function of the smooth control world is zero-free (Remark 5), which is a property of that auxiliary object and is never transferred to the external ordinates. No statement about the position of any zero is drawn from any observable in this paper. Theorem 6.1 states in which sense such an inference is barred: the observables are even in the signed displacement, so its orientation is not recoverable and its magnitude is invisible to first order.

Synthetic worlds are comparison data. The constructed worlds of § 6 are comparison data. No property of any of them is transferred to the zeta function, and the Davenport–Heilbronn data are used as ordinates only.

No claim of an RH criterion. Nothing in this paper is offered as a criterion for, or as evidence for or against, the Riemann Hypothesis. The negativity of the curvature is a one-sided second-moment fact; the classification is a statement about the reach of an instrument.

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Call for Feedback

This paper is part of an open research initiative on prime–zero coupling and explicit-formula operators. All papers in the series are freely available on Zenodo under CC BY 4.0, and the accompanying verification code is hosted on GitHub.

Zenodo (<https://zenodo.org/search?q=Ulrich+Tehrani>) and GitHub (<https://github.com>).

[com/utehrani/analysislab-nt](https://github.com/utehrani/analysislab-nt)).

Topics of particular interest include: the combinatorial question of Open Problem 9.7; sharpenings of the factorisation barrier beyond the bounded-Lipschitz class; and any observable of the coupling data that separates an arithmetic world from a density skeleton under a criterion fixed in advance.

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